

STACKELBERG EQUILIBRIA IN MONOPOLY INSURANCE MARKETS WITH PROBABILITY WEIGHTING

MARIA ANDRAOS
UNIVERSITY OF WATERLOO

MARIO GHOSOUB
UNIVERSITY OF WATERLOO

BIN LI
UNIVERSITY OF WATERLOO

BENXUAN SHI
UNIVERSITY OF WATERLOO

MAY 22, 2026

ABSTRACT. We study Stackelberg Equilibria (Bowley optima) in a monopolistic centralized sequential-move insurance market, with a profit-maximizing insurer who sets premia using a distortion premium principle, and a single policyholder who seeks to minimize a distortion risk measure. We show that equilibria are characterized as follows: In equilibrium, the optimal indemnity function exhibits a layer-type structure, providing full insurance over any loss layer on which the policyholder is more pessimistic than the insurer's pricing functional about tail losses; and no insurance coverage over loss layers on which the policyholder is less pessimistic than the insurer's pricing functional about tail losses. In equilibrium, the optimal pricing distortion function is determined by the policyholder's degree of risk aversion, whereby prices never exceed the policyholder's marginal willingness to insure tail losses. Moreover, we show that both the insurance coverage and the insurer's expected profit increase with the policyholder's degree of risk aversion. Additionally, and echoing recent work in the literature, we show that equilibrium contracts are Pareto efficient, but they do not induce a welfare gain to the policyholder. Conversely, any Pareto-optimal contract that leaves no welfare gain to the policyholder can be obtained as an equilibrium contract. Finally, we consider a few examples of interest that recover some existing results in the literature as special cases of our analysis.

JEL Classification: C61; C62; C72; C79; D86; G22.

Key Words and Phrases: Distortion risk measure; Distortion premium principle; Probability weighting; Stackelberg equilibria; Bowley optima; Pareto optima.

We are grateful to two anonymous reviewers for comments. Mario Ghossoub acknowledges financial support from the Natural Sciences and Engineering Research Council of Canada (NSERC Grant No. 2024-03744). Bin Li acknowledges financial support from (NSERC Grant No. 2020-04338). Benxuan Shi acknowledges financial support from the Society of Actuaries through the Hickman Scholars Program.

1. INTRODUCTION

In monopoly insurance markets under perfect information, the classical literature has been mostly interested in characterizing the insurer's profit-maximizing (or welfare-maximizing) insurance contracts, without any consideration for strategic interaction between the insurer and insured. Whenever such strategic considerations are introduced into the market model, the natural framework is a sequential-move game with the insurer as the leader and the insured as the follower, borrowing from the literature on bilateral monopoly. This is a two-stage game, whereby in the first stage, the insured selects their optimal indemnification given a certain pricing mechanism selected by the insurer; and in the second stage, the insurer observes the insured's demand function and sets prices so as to maximize profit or welfare. The associated strategic equilibrium concept has been termed the *Stackelberg equilibrium*, or *Bowley optimum*.

Bowley optima were first introduced by [Bowley \(1928\)](#) in the context of a bilateral monopoly, and then first applied to insurance markets by [Chan and Gerber \(1985\)](#), in the context of Expected-Utility (EU) preferences with exponential utility functions. Several extensions and/or modifications to this model have subsequently been proposed. For instance, [G.Taylor \(1992\)](#) extends these findings to more general risk exchanges with EU preferences. [Cheung et al. \(2019\)](#) consider the case of markets in which the insurer is a risk-neutral Expected utility maximizer, who sets premia using a distortion premium principle, and a insured who seeks to minimize a distortion risk measure (DRM). They assume that the insured's distortion function is either a concave function (indicating risk aversion), or a distortion function corresponding to the Value-at-Risk (VaR) risk measure. [Li and Young \(2021\)](#) characterize Stackelberg equilibria when agents use a mean-variance functional. [Boonen et al. \(2021\)](#) and [Boonen and Zhang \(2022\)](#) examine the effect of information asymmetry in the context of DRMs. [Ghossoub et al. \(2025\)](#) propose an extension in a different direction, by examining the optimal (nonlinear) pricing mechanisms in the market for the class of deductible and coinsurance indemnity functions. [Boonen and Ghossoub \(2023\)](#) examine the relationship between Bowley optimality and Pareto optimality, under fairly general preferences. [Zhu et al. \(2023\)](#) and [Ghossoub and Zhu \(2024\)](#) provide the first extensions beyond the case of a two-agent insurance market. The former consider a market with multiple insurer having the first move advantage, and one insured; whereas the latter consider the case of one monopoly insurance facing demand from several policyholders. [Andraos et al. \(2026\)](#) recently provided a unification and an extension thereof to more general preferences.

Beyond the static framework discussed above, a growing literature has studied Stackelberg equilibria in dynamic settings using stochastic differential game techniques, in which the reinsurer, acting as the leader, and the insurer, acting as the follower, interact sequentially over time. In particular, [Chen and Shen \(2018\)](#) study a continuous-time Stackelberg reinsurance game in which both parties aim to maximize the expected utility of their terminal wealth. [Chen and Shen \(2019\)](#) extend this framework to a mean-variance setting. More recent contributions extend this framework to settings involving random horizons, ambiguity aversion, and Lévy risk processes. In particular, [Li and Young \(2022\)](#) consider the problem with a random time horizon under the mean-variance principle, while [Cao et al. \(2022\)](#) study a dynamic Stackelberg game between an ambiguity-averse insurance buyer and seller under the spectrally negative Lévy processes. [Cao et al. \(2023b\)](#) extend the model of [Cao et al. \(2022\)](#) to a general divergence penalty for ambiguity. Several extensions also incorporate investment decisions and interactions in stochastic financial markets (e.g., [Gu et al. \(2020\)](#), [Wang and Siu \(2020\)](#), [Bai et al. \(2021\)](#), [Bai et al. \(2022\)](#), [Yang et al. \(2022\)](#)), while more recent studies consider dynamic settings with multiple reinsurers (e.g., [Cao et al. \(2023a\)](#), [Cao et al. \(2024\)](#), [Chen et al. \(2020\)](#)).

In this paper, we consider a monopoly insurance market with a single policyholder. We assume that the policyholder evaluates insurance contracts using a distortion risk measure, following the approach of [Assa \(2015\)](#), [Cheung et al. \(2019\)](#), or [Zhu et al. \(2023\)](#), for instance. The insurer is a risk-neutral profit maximizer, who sets premia using a distortion premium principle. To avoid *ex post* moral hazard that might arise from the policyholder’s misreporting of the true value of the loss, we impose the customary *no sabotage* condition of [Carlier and Dana \(2003, 2005\)](#) on the set of acceptable indemnity functions. Our framework is most closely related to that of [Cheung et al. \(2019\)](#). However, in contrast to their study, which assumes that the policyholder is either strictly risk averse (strictly concave distortion function) or a VaR minimizer, we impose no specific restriction on the curvature of the policyholder’s distortion function. This is a significant extension, as it accommodates for several forms of distortion functions that are empirically more relevant than the concave distortion functions, such as the inverse-S-shaped distortion functions of [Tversky and Kahneman \(1992\)](#), the S-shaped distortion functions of [Prelec \(1998\)](#), or the very flexible class of distortion functions recently introduced by [Bleichrodt et al. \(2023\)](#).

To characterize Stackelberg equilibria in our model, we proceed in two steps. First for a fixed distortion function used to determine the distortion premium principle, we determine the optimal indemnity function that minimizes the policyholder’s risk measure of their end-of-period risk exposure (Theorem 3.6). Second, using the resulting optimal indemnity, we find the pricing distortion function that maximizes the insurer’s expected profit (Corollary 3.10). We show that, in the first step, the optimal indemnity function exhibits a layer-type structure, determined by the interplay between the policyholder’s distortion function and the insurer’s pricing distortion function. Specifically, the optimal indemnity provides full indemnification over any loss layer on which the policyholder is more pessimistic than the insurer’s pricing functional about tail losses. When the policyholder is less pessimistic than the insurer’s pricing functional about tail losses, no indemnification is offered. Finally, when the policyholder and the insurer’s pricing functional are equally pessimistic about tail losses, the marginal indemnity may take an arbitrary shape, within the global 1-Lipschitz (i.e., comonotonicity) constraints. In the second step, the optimal pricing distortion is determined by the policyholder’s degree of (weak) risk aversion, that is, whether the policyholder’s distortion function is above or below the identity function. In equilibrium, the insurer selects a pricing distortion that is aligned with the policyholder’s risk perception, as encoded by their distortion function, in the sense that prices never exceed the policyholder’s marginal willingness to insure tail losses. Moreover, we show that both the insurance coverage and the insurer’s expected profit increase with the policyholder’s degree of strong or weak risk aversion.

We also analyze the Pareto efficiency of the Stackelberg equilibrium contracts. Our results show that any Stackelberg equilibrium contract is Pareto optimal and makes the policyholder indifferent between participating and not participating in the insurance market. Moreover, any Pareto-optimal contract in which the policyholder is indifferent between participation and non-participation can be obtained at a Stackelberg equilibrium. These findings echo similar results obtained by [Boonen and Ghossoub \(2023\)](#) and [Ghossoub and Zhu \(2024\)](#), and they highlight a well-known fundamental phenomenon that occurs in monopoly markets, whereby all consumer surplus is extracted by the monopoly.

When the policyholder is either weakly risk averse (with a distortion function that lies above the identity function) or strongly risk averse (with a concave distortion function), we show that the optimal contract provides full insurance, and the optimal pricing distortion function coincides with that of the policyholder. This recovers the result of [Cheung et al. \(2019\)](#) in the case of concave distortion function. For policyholders who minimize VaR at a confidence level $\alpha \in (0, 1)$, the optimal coverage includes an upper limit. Specifically, the contract provides full insurance for losses

below the α -quantile while leaving the upper tail uninsured. This recovers another result of [Cheung et al. \(2019\)](#). However, in contrast, we show that for individuals with inverse-S shaped distortion functions, reflecting strong sensitivity to extreme losses, the optimal indemnity function takes the form of a deductible contract, whereby extreme losses are fully transferred to the insurer.

The remainder of this paper is organized as follows. Section 2 introduces the insurance market setting, including the risk preferences of the agents, the market mechanisms and resulting insurance contracts, and the (Stackelberg, or Bowley) equilibrium concept. Section 3 provides a characterization of Stackelberg equilibria in our setting. Section 4 examines the Pareto efficiency of the equilibrium contracts, and provides a version of the two welfare theorems. Section 5 presents several examples of interest, for specific types of policyholders. Finally, Section 6 concludes. Proofs and related analysis are given in the [Appendices](#).

2. PROBLEM FORMULATION

Let $B(\mathcal{F})$ denote the set of all bounded random variables on a given non-atomic probability space $(\Omega, \mathcal{F}, \mathbb{P})$. An individual, or potential policyholder, is subject to an insurable loss, which we model as a random variable $X \in B^+(\mathcal{F})$, the positive cone of $B(\mathcal{F})$, with range $[0, M]$. A positive realization of X is seen as a loss. We denote by F_X the cumulative distribution function of X , and by F_X^{-1} the left-continuous inverse of F_X , i.e., the quantile of X , defined as:

$$F_X^{-1}(t) = \inf \{z \in \mathbb{R}^+ \mid F_X(z) \geq t\}, \quad \forall t \in [0, 1].$$

Let $\mathcal{Q}$ denote the set of quantile functions of random variables in $B(\mathcal{F})$. That is,

$$\mathcal{Q} = \{q : (0, 1) \rightarrow \mathbb{R}^+ \mid q \text{ is non-decreasing and left-continuous}\}.$$

2.1. Preferences.

Definition 2.1. A distortion risk measure ρ on $(\Omega, \mathcal{F}, \mathbb{P})$, is defined as the Choquet integral with respect to the distorted probability measure $T \circ \mathbb{P}$. Namely, for any $Y \in B(\mathcal{F})$,

$$\rho(Y) = \int Y \, dT \circ \mathbb{P} = \int_0^{+\infty} T(\mathbb{P}(Y \geq y)) \, dy + \int_{-\infty}^0 [T(\mathbb{P}(Y \geq y)) - 1] \, dy,$$

where $T : [0, 1] \rightarrow [0, 1]$ is a distortion function, that is, a non-decreasing differentiable mapping, satisfying $T(0) = 0$ and $T(1) = 1$.

The conjugate of the distortion function T , is given by: $\tilde{T}(t) = 1 - T(1 - t)$, for all $t \in [0, 1]$. It is easy to verify that $\tilde{T}$ is a distortion function.

The policyholder's preference over $B(\mathcal{F})$ is assumed to admit a distortion risk measure representation, associated with a distortion function T . Specifically, for any $Z \in B(\mathcal{F})$, the policyholder evaluates risk according to

$$\rho^{\text{Pol}}(Z) := \int Z \, dT \circ \mathbb{P}. \quad (1)$$

The induced preference relation $\succsim$ on $B(\mathcal{F})$ is defined by $Z_1 \succsim Z_2$, if and only if, $\rho^{\text{Pol}}(Z_1) \leq \rho^{\text{Pol}}(Z_2)$. Indifference is defined by $Z_1 \sim Z_2$, whenever $Z_1 \succsim Z_2$, and $Z_2 \succsim Z_1$.

2.2. Risk Aversion. Given two random variables $Z_1, Z_2 \in B(\mathcal{F})$, we say that Z_1 dominates Z_2 in *Second-Order Stochastic Dominance* (SSD), written $Z_1 \succ_{SSD} Z_2$, if

$$\int_{-\infty}^x F_{Z_1}(t) dt \leq \int_{-\infty}^x F_{Z_2}(t) dt, \text{ for all } x \in \mathbb{R}.$$

Moreover, Z_2 is said to be a *Mean-Preserving Increase in Risk* (MPIR) of Z_1 if $\mathbb{E}[Z_1] = \mathbb{E}[Z_2]$ and $Z_1 \succ_{SSD} Z_2$. We next discuss weak and strong risk aversion of a preference $\succsim$ over $B(\mathcal{F})$.

Definition 2.2. A preference $\succsim$ over $B(\mathcal{F})$ is said to be *weakly risk averse*, if $\mathbb{E}[Z] \succsim Z$ for all $Z \in B(\mathcal{F})$.

Definition 2.3. A preference $\succsim$ over $B(\mathcal{F})$ is said to be *strongly risk averse* if it ranks any random variable above all of its mean-preserving increases in risk. That is, $\succsim$ is strongly risk averse if $Z_1 \succ Z_2$, for all $Z_1, Z_2 \in B(\mathcal{F})$, such that Z_2 is a MPIR of Z_1 .

Strong risk aversion implies weak risk aversion, since any $Z \in B(\mathcal{F})$ is a mean-preserving increase in risk of $\mathbb{E}[Z]$. Moreover, by the positive homogeneity of the Choquet integral, it follows that,

$$\mathbb{E}[Z] \succsim Z \iff \mathbb{E}[Z] \leq \rho^{Pol}(Z), \forall Z \in B(\mathcal{F}).$$

By a classical result (see, e.g., Yaari (1987)), risk aversion under distortion risk measures is characterized by properties of the distortion function T . In particular, the policyholder is weakly risk averse if and only if $T(t) \geq t$ for all $t \in [0, 1]$. Moreover, the policyholder is strongly risk averse if and only if the distortion function T is concave.

Recall that for a given preference relation $\succsim$, the certainty equivalent of $Z \in B(\mathcal{F})$ is the constant $CE^\succsim(Z) \in \mathbb{R}$ such that

$$Z \sim CE^\succsim(Z),$$

and the risk premium associated with $Z \in B(\mathcal{F})$ is defined as

$$\Delta^\succsim(Z) := \mathbb{E}[Z] - CE^\succsim(Z).$$

Following Chew et al. (1987), Quiggin (1993), and Ghossoub and He (2021), we define comparative notions of risk aversion below.

Definition 2.4 (Comparative risk aversion). Consider two preference relations $\succsim$ and $\succsim^*$ over $B(\mathcal{F})$:

- (1) $\succsim^*$ is said to be *more weakly risk averse* than $\succsim$ if, for any $Z \in B(\mathcal{F})$, $\Delta^{\succsim^*}(Z) \geq \Delta^\succsim(Z)$.
- (2) $\succsim^*$ is said to be *more strongly risk averse* than $\succsim$ if $Z_1 \succsim^* Z_2$ for any $Z_1, Z_2 \in B(\mathcal{F})$ such that:
 - (a) $Z_1 \sim Z_2$.
 - (b) There exists $z_0 \in \mathbb{R}$ with $F_{Z_2}(z) \geq F_{Z_1}(z)$ for all $z < z_0$, and $F_{Z_2}(z) \leq F_{Z_1}(z)$ for all $z \geq z_0$.

By a classical result (e.g., Chew et al. (1987)), we obtain the following characterization of comparative weak and strong risk aversion for distortion risk measures.

Proposition 2.5. *Consider two policyholders whose preferences $\succsim$ and $\succsim^*$ over $B(\mathcal{F})$ admit representations by distortion risk measures ρ^{Pol} and ρ^{*Pol} respectively. Let T and T^* denote the respective distortion functions of each policyholder. Then the following holds:*

- (1) *The second policyholder is more weakly risk averse than the first policyholder if and only if $T^*(t) \geq T(t)$, for all $t \in [0, 1]$.*
- (2) *The second policyholder is more strongly risk averse than the first policyholder if and only if T^* is a concave transformation of T . That is, there exists an increasing and concave function $g : [0, 1] \rightarrow [0, 1]$, satisfying $g(0) = 0$, $g(1) = 1$, and $T^*(t) = g(T(t))$ for all $t \in [0, 1]$.*

Remark 2.6. *If g is an increasing and concave function on $[0, 1]$ such that $g(0) = 0$ and $g(1) = 1$, then $g(t) \geq t$, for all $t \in [0, 1]$. Consequently, if $T^* \equiv g \circ T$, we obtain that $T^*(t) \geq T(t)$ for all $t \in [0, 1]$. That is, if T^* is more strongly risk averse than T , then T^* is also more weakly risk averse than T .*

2.3. Market Mechanisms and Contracts. The market allows the policyholder to cede part of the loss X to an insurer, in exchange for a premium payment. We assume that the market only offers indemnities in the set of *ex ante* admissible indemnity schedules $\mathcal{I}_L$ defined below.

$$\mathcal{I}_L = \left\{ I : [0, M] \rightarrow [0, M] \mid I(0) = 0, 0 \leq I(x_1) - I(x_2) \leq x_1 - x_2, \forall x_2 \leq x_1 \in [0, M] \right\}.$$

That is, $\mathcal{I}_L$ is the set of 1-Lipschitz functions satisfying the so-called *no-sabotage condition* of [Carlier and Dana \(2003\)](#), so as to rule *ex post* moral hazard that could arise from misreporting of the actual realized loss.

Definition 2.7. *An insurance contract is a pair (I, π) , where $I \in \mathcal{I}_L$ is an indemnity function and $\pi \in \mathbb{R}$ is the premium paid by the policyholder for coverage I .*

Assumption 2.8. *The insurer is assumed to price insurance using a distortion premium principle of the form:*

$$\Pi_g(I(X)) := \int I(X) dg \circ \mathbb{P}, \quad \forall I \in \mathcal{I}_L, \quad (2)$$

for some distortion function g , which we hereafter refer to as the pricing distortion used by the insurer.

Definition 2.9. *A market mechanism is a pair (I, g) , where $I \in \mathcal{I}_L$ is an indemnity function and g is a pricing distortion function.*

A market mechanism (I, g) induces an insurance contract of the form $(I, \Pi_g(I(X)))$, where the premium is computed using the pricing distortion g . The insurer's end-of-period profit is therefore given by $\Pi_g(I(X)) - I(X)$, while the policyholder's end-of-period risk exposure is given by $X - I(X) + \Pi_g(I(X))$. Accordingly, the insurer's resulting expected profit is:

$$V^{In}(I, g) = \Pi_g(I(X)) - \mathbb{E}[I(X)], \quad (3)$$

and the policyholder evaluates this risk exposure using:

$$\rho^{Pol}(I, g) = \rho^{Pol}\left(X - I(X) + \Pi_g(I(X))\right). \quad (4)$$

Letting $R(X) := X - I(X)$ denote the retention function, i.e., the part of the loss X that is retained by the policyholder, and by translation invariance of ρ^{Pol} , (4) reduces to:

$$\rho^{Pol}(I, g) = \rho^{Pol}(R(X)) + \Pi_g(I(X)). \quad (5)$$

Remark 2.10. An indemnity function I belongs to $\mathcal{I}_L$ if and only if the corresponding retention function $R(X) = X - I(X)$ belongs to $\mathcal{I}_L$. Moreover, since I is non-decreasing, the random variables $I(X)$ and $R(X)$ are comonotonic¹.

2.4. A Sequential-Move Game. The insurance market is modeled as a sequential move game, in which the insurer, having the first-mover advantage, starts by selecting a pricing distortion function g . Given that choice, the policyholder then selects an indemnity function that minimizes their risk exposure $\rho^{Pol}(I, g)$. Anticipating the policyholder's optimal indemnity choice as a function of the selected pricing distortion function g , the insurer selects the optimal distortion function g^* that maximizes their expected profit $V^{In}(I, g)$. The equilibrium concept that is best suited for this sequential game is the Stackelberg equilibrium.

Definition 2.11. A given market mechanism (I^*, g^*) is said to be a Stackelberg Equilibrium (SE), if

- (1) $I^* \in \arg \min_{I \in \mathcal{I}_L} \rho^{Pol}(I, g^*)$, and
- (2) $V^{In}(I^*, g^*) \geq V^{In}(I, g)$, for all (I, g) such that $I \in \arg \min_{I \in \mathcal{I}_L} \rho^{Pol}(\bar{I}, g)$.

Definition 2.11 suggests that Stackelberg equilibria can be characterized through a two-step procedure. In the first step, for a fixed pricing distortion function g , the policyholder chooses an indemnity function that minimizes their risk exposure. This problem will be referred to as the *policyholder's problem*. In the second step, anticipating the policyholder's optimal response as a function of g , the insurer selects a pricing distortion function g^* that maximizes expected profit. This problem will be referred to as the *insurer's problem*.

For a given insurance contract $(I, \pi) \in \mathcal{I}_L \times \mathbb{R}$, the policyholder's risk exposure and the insurer's expected profit can be written as

$$\rho^{Pol}(I, \pi) = \rho^{Pol}(X - I(X) + \pi) \quad \text{and} \quad V^{In}(I, \pi) = \pi - \mathbb{E}[I(X)].$$

Definition 2.12. An insurance contract $(I^*, \pi^*) \in \mathcal{I}_L \times \mathbb{R}$ is said to be individually rational, IR, if

$$\rho^{Pol}(I^*, \pi^*) \leq \rho^{Pol}(0, 0) \quad \text{and} \quad V^{In}(I^*, \pi^*) \geq V^{In}(0, 0).$$

Definition 2.12 states that an insurance contract $(I^*, \pi^*) \in \mathcal{I}_L \times \mathbb{R}$ is individually rational, if it incentivizes the policyholder and the monopolist insurer to participate in the market.

Definition 2.13. An insurance contract $(I^*, \pi^*) \in \mathcal{I}_L \times \mathbb{R}$ is said to be Pareto optimal, PO, if there does not exist another contract $(I, \pi) \in \mathcal{I}_L \times \mathbb{R}$ such that

$$\rho^{Pol}(I, \pi) \leq \rho^{Pol}(I^*, \pi^*) \quad \text{and} \quad V^{In}(I, \pi) \geq V^{In}(I^*, \pi^*),$$

with at least one strict inequality.

¹Two random variables $X, Y \in B(\mathcal{F})$ are said to be comonotone if $(X(\omega_1) - X(\omega_2))(Y(\omega_1) - Y(\omega_2)) \geq 0$, for all $\omega_1, \omega_2 \in \Omega$.

Definition 2.13 states that an insurance contract $(I^*, \pi^*) \in \mathcal{I}_L \times \mathbb{R}$ is Pareto optimal if there is no alternative contract that weakly reduces the policyholder's risk exposure and weakly increases the insurer's profit, with at least one of these improvements being strict.

3. CHARACTERIZATION OF STACKELBERG EQUILIBRIA

In this Section, we aim to characterize Stackelberg equilibria through a two-step procedure as suggested in Definition 2.11. Specifically, in Subsection 3.1, we study the policyholder's problem, which constitutes the first step in determining Stackelberg equilibria. Then, in Subsection 3.2, we consider the second step by addressing the insurer's problem.

3.1. The Policyholder's Problem. For a given choice of pricing distortion function g , the policyholder chooses an indemnity $I \in \mathcal{I}_L$ to minimize risk exposure:

$$\min_{I \in \mathcal{I}_L} \rho^{Pol}(I, g). \quad (6)$$

To analyze the policyholder's problem given in (6), we impose the following assumption on the loss distribution that will allow us to reformulate the problem in terms of quantile functions.

Assumption 3.1. *The cumulative distribution function F_X is strictly increasing.*

It follows from Assumption 3.1 that F_X is differentiable almost everywhere. Moreover, by Föllmer and Schied (2016, Lemma A.25), Assumption 3.1 also guarantees that $U := F_X(X)$ is uniformly distributed on $(0, 1)$, and that $X = F_X^{-1}(U)$, $\mathbb{P}$ -a.s.

Remark 3.2. *For each $I \in \mathcal{I}_L$, $I(X)$ and $X - I(X)$ have strictly increasing cumulative distribution functions by Assumption 3.1. Consequently, their quantile functions are strictly increasing, left-continuous, and differentiable a.e. on $[0, 1]$.*

For any $Z \in B^+(\mathcal{F})$ whose quantile function F_Z^{-1} is differentiable almost everywhere, the policyholder's risk measure admits the following representation:

$$\rho^{Pol}(Z) = \int Z dT \circ \mathbb{P} = \int (F_Z^{-1})'(U) T(1 - U) d\mathbb{P}.$$

Similarly, the premium evaluated under the given pricing distortion g can be written as

$$\Pi_g(Z) = \int Z dg \circ \mathbb{P} = \int (F_Z^{-1})'(U) g(1 - U) d\mathbb{P}.$$

Applying these formulas to the retention $R(X)$ and the indemnity $I(X)$ respectively, and using expression (5) of $\rho^{Pol}(I, g)$, we obtain

$$\rho^{Pol}(I, g) = \int (F_{R(X)}^{-1})'(U) T(1 - U) d\mathbb{P} + \int (F_{I(X)}^{-1})'(U) g(1 - U) d\mathbb{P}. \quad (7)$$

Remark 3.3. *Since $X = I(X) + R(X)$, we can write $F_X^{-1} = F_{I(X)}^{-1} + F_{R(X)}^{-1}$, by comonotonic additivity of the quantile function.*

As a result of the above remark, (7) can be rewritten as follows:

$$\rho^{Pol}(I, g) = \int_0^1 (F_{R(X)}^{-1})'(t) [T(1-t) - g(1-t)] dt + \int_0^1 (F_X^{-1})'(t) g(1-t) dt.$$

Let $\mathcal{Q}_L$ be the set of admissible quantile functions, defined as

$$\mathcal{Q}_L = \left\{ q \in \mathcal{Q} \mid q(0) = 0, 0 \leq q'(t) \leq (F_X^{-1})'(t) \right\}.$$

Each $q \in \mathcal{Q}_L$ corresponds to the quantile of the retention random variable $R(X) = X - I(X)$ for some admissible $I \in \mathcal{I}_L$. Hence, for a given $q \in \mathcal{Q}_L$, we have:

$$\rho^{Pol}(q, g) = \int_0^1 q'(t) [T(1-t) - g(1-t)] dt + \int_0^1 (F_X^{-1})'(t) g(1-t) dt. \quad (8)$$

Reformulating the policyholder's problem (6) in quantile form, we obtain:

$$\min_{q \in \mathcal{Q}_L} \rho^{Pol}(q, g). \quad (9)$$

Lemma 3.4. *For a given pricing distortion function g , the feasible quantile $q_g(t) \in \mathcal{Q}_L$ is optimal for (9) if and only if the indemnity $I_g(x) = x - q_g(F_X(x))$ is optimal for (6).*

Proof. The proof can be found in Appendix A.1. □

Lemma 3.5. *For a given pricing distortion function g , a quantile function q_g is optimal for (9) if and only if*

$$(q_g)'(t) = \begin{cases} 0, & g(1-t) < T(1-t), \\ \phi_g(t), & g(1-t) = T(1-t), \\ (F_X^{-1})'(t), & g(1-t) > T(1-t), \end{cases} \quad (10)$$

where $\phi_g(t) \in [0, (F_X^{-1})'(t)]$, for almost every $t \in [0, 1]$ such that $g(1-t) = T(1-t)$.

Proof. The proof can be found in Appendix A.2. □

Theorem 3.6. *For a given pricing distortion function g , an indemnity function I_g is optimal for the policyholder's problem given in (6) if and only if it is of the form $I_g(x) = \int_0^x \kappa(y) dy$, for all $x \in [0, M]$, with $\kappa : [0, M] \rightarrow [0, 1]$ satisfying the following:*

$$\kappa(y) = \begin{cases} 1, & g(\mathbb{P}[X > y]) < T(\mathbb{P}[X > y]), \\ 1 - \phi_g(F_X(y)) f_X(y), & g(\mathbb{P}[X > y]) = T(\mathbb{P}[X > y]), \\ 0, & g(\mathbb{P}[X > y]) > T(\mathbb{P}[X > y]), \end{cases}$$

where $\phi_g(t) \in [0, (F_X^{-1})'(t)]$, for almost every $t \in [0, 1]$ such that $g(1-t) = T(1-t)$, and f_X denotes the probability density function of the loss X .

Proof. The proof follows immediately from Lemmata 3.4 and 3.5. □

Theorem 3.6 characterizes the set of optimal indemnity functions for a given pricing distortion function g , in terms of the policyholder's marginal indemnification. Specifically, full indemnification is optimal when the insurer's pricing distortion assigns less weight to tail probabilities than the policyholder's distortion function T . The policyholder retains the entire loss if the insurer's pricing distortion overweights the tail probability compared to the policyholder's distortion. Finally, when the policyholder's distortion is equal to the insurer's pricing distortion at a given tail probability, the policyholder may receive partial coverage, as long as feasibility is maintained.

This structural characterization is consistent with Assa (2015), who considers a reinsurance problem in which the premium principle is fixed and distortion based, and characterizes the optimal contract of the policyholder. In contrast, our result arises as the policyholder's best response within a Stackelberg framework, where the pricing distortion is a strategic choice of the insurer. Moreover, in the absence of strategic interaction and when the policyholder's problem only is considered, our model reduces to the setting analyzed in Assa (2015).

3.2. The Insurer's Problem. The optimal indemnity characterized in Theorem 3.6 is not unique. The insurer's objective is to identify a market mechanism (I_g^*, g^*) that solves the following problem:

$$\max_g V^{In}(I_g, g), \text{ such that } I_g \in \arg \min_{I \in \mathcal{I}_L} \rho^{Pol}(I, g). \quad (11)$$

Lemma 3.7. *The market mechanism (I_g^*, g^*) is a Stackelberg equilibrium if and only if it is optimal for the insurer's problem in (11).*

Proof. The proof follows immediately from Definition 2.11. □

The insurer's problem (11) can be reformulated using quantile functions, similarly to the policyholder's problem analyzed in Subsection 3.1. Consider a market mechanism (I, g) , using Remark 3.3, the premium can be written as

$$\Pi_g(I(X)) = \int_0^1 \left[(F_X^{-1})'(t) - (F_{R(X)}^{-1})'(t) \right] g(1-t) dt.$$

Moreover,

$$\mathbb{E}[I(X)] = \int_0^1 \left[F_X^{-1}(t) - F_{R(X)}^{-1}(t) \right] dt.$$

Substituting the quantile representations of the premium and the expected indemnity into $V^{In}(I_g, g)$ in (3), where R_g denotes the retention function associated with I_g , yields:

$$V^{In}(I_g, g) = \int_0^1 \left[(F_X^{-1})'(t) - (F_{R_g(X)}^{-1})'(t) \right] g(1-t) dt - \int_0^1 \left[F_X^{-1}(t) - F_{R_g(X)}^{-1}(t) \right] dt.$$

For a given quantile $q \in \mathcal{Q}_L$, this expression reduces to:

$$V^{In}(q, g) = \int_0^1 \left[(F_X^{-1})'(t) - q'(t) \right] g(1-t) dt - \int_0^1 \left[F_X^{-1}(t) - q(t) \right] dt.$$

Hence, the insurer's problem can be equivalently written in quantile form as follows:

$$\max_g V^{In}(q_g, g), \text{ such that } q_g \in \arg \min_{q \in \mathcal{Q}_L} \rho^{Pol}(q, g). \quad (12)$$

The following result suggests that solving the insurer's problem (11) is equivalent to solving (12), providing a one-to-one correspondence between optimal solutions in the quantile space and optimal mechanisms.

Lemma 3.8. (q_g, g) is optimal for (12) if and only if (I_g, g) is optimal for (11), where $I_g(x) = x - q_g(F_X(x))$, for all $x \in [0, M]$.

Proof. The proof is similar to that of Lemma 3.4. $\square$

Theorem 3.9. $(q_{g^*}^*, g^*)$ is optimal for (12) if and only if the following two conditions hold:

- (1) The optimal pricing distortion g^* satisfies $g^*(t) = 1 - \tilde{g}^*(1 - t)$ for all $t \in [0, 1]$, where the optimal pricing conjugate $\tilde{g}^*$ is given by:

$$\tilde{g}^*(t) = \begin{cases} \tilde{T}(t), & \tilde{T}(t) < t, \\ \in [\sup \{z < t; \tilde{g}^*(z)\}, \tilde{T}(t)], & \tilde{T}(t) \geq t, \end{cases} \quad (13)$$

and $\tilde{T}$ is the conjugate distortion function of T .

- (2) The optimal quantile $q_{g^*}^*$ satisfies:

$$(q_{g^*}^*)'(t) = \begin{cases} 0, & \tilde{T}(t) < t, \\ \phi_{g^*}(t) & t \in \{z; \tilde{g}^*(z) = \tilde{T}(z) = z\}, \\ (F_X^{-1})'(t), & t \in \{z; \tilde{g}^*(z) < \tilde{T}(z) = z\} \cup \{z; \tilde{T}(z) > z\}, \end{cases} \quad (14)$$

where $\phi_{g^*}(t) \in [0, (F_X^{-1})'(t)]$ for a.e. $t \in [0, 1]$ such that $\tilde{g}^*(t) = \tilde{T}(t) = t$.

Moreover, the insurer's expected profit under $(q_{g^*}^*, g^*)$ is given by:

$$V^{In}(q_{g^*}^*, g^*) = \int_0^1 (F_X^{-1})'(t) (t - \tilde{T}(t)) \mathbb{1}_{\{\tilde{T}(t) < t\}} dt. \quad (15)$$

Proof. The proof can be found in Appendix A.3. $\square$

Theorem 3.9 provides a complete characterization of the Stackelberg equilibrium by identifying the optimal pricing distortion g^* and the induced optimal quantile $q_{g^*}^*$. The following corollary explicitly provides the characterization of Stackelberg equilibria in terms of market mechanisms of the form (I_g, g) .

Corollary 3.10. The market mechanism $(I_{g^*}^*, g^*)$ is optimal for (11) if and only if the following two conditions hold:

- (1) The optimal pricing distortion g^* is given by:

$$g^*(t) = \begin{cases} T(t), & T(t) > t, \\ \in [T(t), \inf \{z > t : g^*(z)\}], & T(t) \leq t, \end{cases} \quad (16)$$

(2) The optimal indemnity satisfies $I_{g^*}^*(x) = \int_0^x \kappa^*(y) dy$ for all $x \in [0, M]$, where $\kappa^* : [0, M] \rightarrow [0, 1]$ is defined as follows:

$$\kappa^*(y) := \begin{cases} 1, & \mathbb{P}[X > y] < T(\mathbb{P}[X > y]), \\ 1 - \phi_{g^*}(F_X(y)) f_X(y), & g^*(\mathbb{P}[X > y]) = T(\mathbb{P}[X > y]) = \mathbb{P}[X > y], \\ 0, & g^*(\mathbb{P}[X > y]) > T(\mathbb{P}[X > y]) = \mathbb{P}[X > y], \text{ or } \mathbb{P}[X > y] > T(\mathbb{P}[X > y]), \end{cases}$$

where $\phi_{g^*}(t) \in [0, (F_X^{-1})'(t)]$, for a.e. $t \in [0, 1]$ such that $g^*(1 - t) = T(1 - t) = 1 - t$.

Proof. The proof follows immediately from Lemma 3.8 and Theorem 3.9. $\square$

Theorem 3.11. Consider two policyholders whose respective distortion functions T_1 and T_2 satisfy $T_1(t) \leq T_2(t)$ for all $t \in [0, 1]$. Let $(I_{g_1^*}^*, g_1^*)$ and $(I_{g_2^*}^*, g_2^*)$ denote the corresponding Stackelberg equilibria. Then, the following holds.

(1) $I_{g_1^*}^*(x) \leq I_{g_2^*}^*(x)$, for all $x \in [0, M]$.

(2) $V^{In}(I_{g_1^*}^*, g_1^*) \leq V^{In}(I_{g_2^*}^*, g_2^*)$

Theorem 3.11 states that under Stackelberg equilibria, the optimal insurance coverage and the insurer's expected profit both increase as the policyholder becomes weakly more risk averse in the sense of Proposition 2.5. That is, the more weakly risk averse the policyholder is, the more coverage they receive, and the more profitable the insurer becomes. The following corollary shows that this result also holds under strong risk aversion.

Corollary 3.12. Under Stackelberg equilibria, if the policyholder becomes more strongly risk averse, then both the optimal insurance coverage and the insurer's expected profit increase.

Proof. The proof of this result follows immediately from Remark 2.6 and Theorem 3.11. $\square$

In contrast to this clear monotonic relationship with respect to risk aversion obtained in Theorem 3.11 and Corollary 3.12, no such monotonicity can generally be established for the insurance coverage or the insurer's expected profit when the policyholder's risk distribution itself changes.

4. PARETO EFFICIENCY OF STACKELBERG EQUILIBRIA

In this section, we examine whether the Stackelberg equilibria characterized in Section 3 lead to Pareto efficient contracts. We first characterize Pareto optimal insurance contracts and then establish their relationship with Stackelberg equilibria. The proofs of these results can be found in Appendix A.5, Appendix A.6, and Appendix A.7.

Proposition 4.1. An insurance contract (I^*, π^*) is Pareto optimal if and only if it solves the following problem:

$$\min_{(I, \pi) \in \mathcal{I}_L \times \mathbb{R}} \left\{ \rho^{Pol}(I, \pi) - V^{In}(I, \pi) \right\}. \quad (17)$$

Proposition 4.2. Let $(I_{g^*}^*, g^*)$ be a Stackelberg equilibrium characterized in Corollary 3.10. The induced insurance contract $(I_{g^*}^*, \pi_{g^*}^*)$ is individually rational and Pareto optimal. Moreover, $\rho^{Pol}(I_{g^*}^*, \pi_{g^*}^*) = \rho^{Pol}(0, 0)$.

Proposition 4.2 establishes a first welfare theorem for the sequential-move monopolistic market considered in this paper. Specifically, Stackelberg equilibria lead to Pareto optimal insurance contracts. Moreover, in equilibrium, the policyholder is indifferent between participating in the market and not participating, and the monopolist insurer extracts all surplus. In contrast, not every Pareto optimal contract arises from a Stackelberg equilibrium mechanism. The following result discusses the conditions under which Pareto optimal contracts are induced by Stackelberg equilibria.

Proposition 4.3. *Consider a Pareto optimal contract (I^*, π^*) satisfying $\rho^{Pol}(I^*, \pi^*) = \rho^{Pol}(0, 0)$. Then there exists a Stackelberg equilibrium $(I^*, g_{I^*}^*)$ that induces the Pareto optimal contract (I^*, π^*) , that is,*

$$\Pi_{g_{I^*}^*}(I^*(X)) = \pi^*.$$

Proposition 4.3 states that the Pareto optimal contracts that leave the policyholder indifferent between participating and not participating in the market are induced by Stackelberg equilibrium mechanisms.

The results of this section are consistent with findings of the existing literature, e.g., Boonen and Ghossoub (2023), Ghossoub and Zhu (2024), in which the monopolistic market is modeled as a sequential-move game. Additionally, Andraos et al. (2026) consider a generalized framework and obtain similar results for the special case of monopoly. Specifically, Stackelberg equilibria yield Pareto optimality without inducing any welfare gain to the policyholder. Conversely, only the Pareto optimal contracts that leave the policyholder indifferent between suffering loss and entering the market are induced from Stackelberg equilibrium mechanisms.

5. EXAMPLES

In this section, we assume that the policyholder evaluates risk using a specific class of risk measures, and we examine the resulting Stackelberg equilibria. As established in Lemma 3.5 and Theorem 3.9, the solutions to both stages of the Stackelberg game are generally not unique. Nevertheless, we show that under certain assumptions, Stackelberg equilibria may take one of several familiar forms such as full insurance, a coverage limit insurance contract, or a deductible insurance contract.

5.1. Optimality of Full Insurance. Suppose that the policyholder is weakly risk averse, that is, the distortion function satisfies

$$T(t) \geq t, \quad \forall t \in [0, 1].$$

It follows from Corollary 3.10, that the market mechanism $(I_{g^*}^*, g^*)$ described below is a Stackelberg equilibrium:

$$g^*(t) = T(t), \quad \forall t \in [0, 1], \quad \text{and} \quad I_{g^*}^*(x) = x, \quad \forall x \in [0, M].$$

That is, the insurer offers full coverage, and the pricing distortion function coincides with the distortion function that reflects the policyholder's risk aversion. This result also holds for a strongly risk averse policyholder, by Remark 2.6.

This example covers many important risk preference models commonly used in practice, such as Tail Value-at-Risk (TVaR), which corresponds to a concave distortion function of the form

$$T(t) := \min\left(1, \frac{t}{1 - \alpha}\right), \quad \text{for a given } \alpha \in (0, 1).$$

Moreover, under TVaR, the insurer's expected profit satisfies:

$$V^{In}(I_{g^*}^*, g^*) = \int_0^1 (F_X^{-1})'(t) [t - \tilde{T}(t)] \mathbb{1}_{\{\tilde{T}(t) < t\}} dt \geq 0,$$

since the set $\{t \in [0, 1] : \tilde{T}(t) < t\}$ has positive measure.

5.2. Optimality of Coverage Limit Contracts. In this example, we assume that the policyholder evaluates risk using Value-at-Risk (VaR) at confidence level $\alpha \in (0, 1)$. This corresponds to a distortion risk measure with distortion function

$$T(t) := \mathbb{1}_{(1-\alpha, 1]}(t), \quad \forall t \in [0, 1].$$

It follows from Corollary 3.10 that a Stackelberg equilibrium $(I_{g^*}^*, g^*)$ is characterized as follows. The optimal pricing distortions g^* satisfies

$$g^*(t) = \begin{cases} 1, & t > 1 - \alpha, \\ \in [0, \inf \{z > t : g^*(z)\}], & t \leq 1 - \alpha, \end{cases} \quad \text{for a.e } t \in [0, 1],$$

and the optimal indemnity function $I_{g^*}^*$ is given by

$$I_{g^*}^*(x) = \begin{cases} x, & x < F_X^{-1}(\alpha), \\ F_X^{-1}(\alpha), & x \geq F_X^{-1}(\alpha), \end{cases} \quad \forall x \in [0, M].$$

In this case, the policyholder receives full coverage for losses below the VaR threshold $F_X^{-1}(\alpha)$. Moreover, coverage is capped at this threshold level, so that for losses exceeding $F_X^{-1}(\alpha)$, the policyholder retains the excess loss. The insurer's expected profit satisfies:

$$V^{In}(I_{g^*}^*, g^*) = \int_0^1 (F_X^{-1})'(t) [t - \tilde{T}(t)] \mathbb{1}_{\{\tilde{T}(t) < t\}} dt = \int_0^\alpha t \cdot (F_X^{-1})'(t) dt,$$

since $\tilde{T}(t) = \mathbb{1}_{[\alpha, 1]}(t)$, for all $t \in [0, 1]$. Additionally, note that since F_X^{-1} is strictly increasing, it follows that for a fixed $x \in [0, M]$, the optimal indemnity function $I_{g^*}^*$ weakly increases with the VaR confidence level α . Moreover, the insurer's expected profit increases with α .

Hence, if the policyholder is a VaR minimizer, then the coverage limit contract is a Stackelberg equilibrium and depends on the chosen VaR confidence level α . In addition, a more risk-averse policyholder (with a higher value for α) receives greater coverage and generates higher expected profit for the insurer, consistently with Theorem 3.11 and Corollary 3.12.

The results of our examples so far, align with the findings of Cheung et al. (2019), where the policyholder is assumed to be either strongly risk averse or a VaR minimizer. Specifically, if the policyholder's preferences are represented by a DRM with a concave distortion function, the results of Example 5.1 align with Cheung et al. (2019, Theorem 3.1, case $\gamma = 0$). On the other hand, if the distortion function corresponds to a VaR risk measure, then the results of Example 5.2 are consistent with Cheung et al. (2019, Theorem 3.5, case $\gamma = 0$).

5.3. Optimality of Deductible Contracts. We now assume that the policyholder uses an inverse-S-shaped distortion function (ISSD), which is commonly used to study decision making under uncertainty (e.g., [Tversky and Kahneman \(1992\)](#)).

Definition 5.1. *A distortion function T is said to be an inverse-S-shaped distortion function (ISSD) if it is twice-differentiable on $(0, 1)$, and there exists $t_0 \in (0, 1)$ such that:*

- (1) $T'(t)$ is strictly decreasing on $(0, t_0)$, and
- (2) $T'(t)$ strictly increasing on $(t_0, 1)$.

Moreover, $\lim_{t \downarrow 0} T'(t) > 1$, and $\lim_{t \uparrow 1} T'(t) > 1$.

In this case, for $Y \in B^+(\mathcal{F})$, the expression of the distortion risk measure is given by

$$\rho(Y) = \int_0^{+\infty} T(\mathbb{P}(Y \geq y)) dy = \int_0^{+\infty} y T'(1 - F_Y(y)) dF_Y(y).$$

Assume that there exists $t_1 \in (0, 1)$ such that $T(t_1) = t_1$. That is, t_1 is the intersection point between the identity function and the policyholder's distortion function. We know from [Corollary 3.10](#), that the Stackelberg equilibrium $(I_{g^*}^*, g^*)$ is characterized as follows. The optimal pricing distortion g^* satisfies:

$$g^*(t) = \begin{cases} T(t), & t < t_1, \\ \in [T(t), \inf\{z > t : g^*(z)\}], & t \geq t_1, \end{cases} \quad \text{for a.e. } t \in [0, 1],$$

and the optimal indemnity function $I_{g^*}^*$ satisfies:

$$I_{g^*}^*(x) = \begin{cases} 0, & x \leq F_X^{-1}(1 - t_1), \\ x - F_X^{-1}(1 - t_1), & x > F_X^{-1}(1 - t_1), \end{cases} \quad \forall x \in [0, M].$$

In this case, the optimal indemnity fully covers losses above a fixed deductible $F_X^{-1}(1 - t_1)$, which is fully determined by t_1 . Hence, if the policyholder is more concerned about extreme losses, then the deductible contract is a Stackelberg equilibrium. Moreover, the insurer's expected profit is given by:

$$V^{In}(I_{g^*}^*, g^*) = \int_0^{t_1} (F_X^{-1})'(1 - t) (T(t) - t) dt.$$

We note that for a fixed $x \in [0, M]$, as t_1 increases, $F_X^{-1}(1 - t_1)$ decreases, implying a greater insurance coverage. That is, the optimal indemnity $I_{g^*}^*$ increases with t_1 . However, from the expression of the expected profit at equilibrium, we cannot conclude whether $V^{In}(I_{g^*}^*, g^*)$ increases with t_1 . This is because a higher value of t_1 implies a change in the distortion function T on the interval $[0, t_1)$. Hence, to evaluate the impact of t_1 on the insurer's expected profit, we consider the following concrete numerical example, in which the distortion function takes the form given in [Tversky and Kahneman \(1992\)](#):

$$T_\theta(t) := \frac{t^\theta}{(t^\theta + (1 - t)^\theta)^{\frac{1}{\theta}}},$$

where $\theta \in [0.3, 0.8]$. Additionally, we consider three cases for the distribution of the random loss X .

We first assume that the random loss X follows a uniform distribution with cumulative distribution function (CDF) given by

$$F_X(x) = \frac{x}{M}, \quad \forall x \in [0, M] \text{ and } M = 10.$$

Figure 1a plots the distortion functions T_θ for different values of θ . The figure shows that as θ increases, T_θ approaches the identity function, and the intersection point t_1 shifts to larger values. Moreover, functions with smaller values of θ are more concave near 0 and more convex near 1, implying that the policyholder places greater emphasis on extreme losses.

Figure 1b depicts the insurer's expected profit under the Stackelberg optimal contract as a function of t_1 . The result shows that the insurer's expected profit does not vary monotonically with θ or the intersection point t_1 . Notably, it reaches a maximum around $t_1 \approx 0.32$.

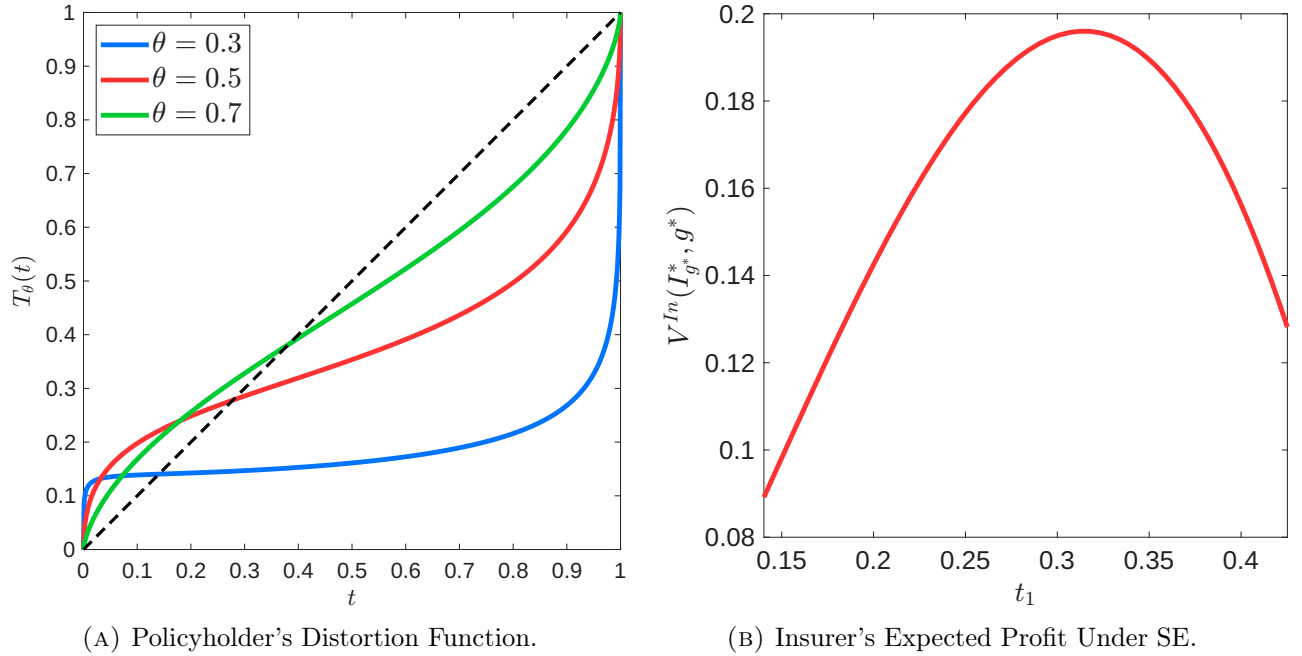


FIGURE 1. The case where X follows a uniform distribution.

Alternatively, assume now that the loss random variable X follows a truncated exponential distribution, with CDF given by

$$F_X(x) = \frac{1 - \exp(-\lambda x)}{1 - \exp(-\lambda M)}, \quad \forall x \in [0, M], \quad M = 10, \text{ and } \lambda > 0.$$

Figure 2a plots the CDF for different values of λ . It shows that as λ increases, the CDF becomes steeper for small losses, implying that losses are more likely to take smaller values. Figure 2b plots the insurer's profit as a function of t_1 for different values of λ . The insurer's profit does not vary monotonically with the parameter θ or the intersection point t_1 .

The result also depends on the loss distribution parameter λ . In particular, the insurer's profit reaches its maximum around $t_1 \approx 0.3$ when $\lambda = 0.1$. When λ increases to 0.5, meaning that the distribution becomes more concentrated on the left, the maximum shifts to about $t_1 \approx 0.22$,

indicating that if the policyholder faces a lower probability of large losses, greater concern for extreme losses becomes more valuable to the insurer. When λ increases further to 1, implying that losses are even more likely to be small, the maximum shifts slightly further left, and the insurer's profit then decreases monotonically with t_1 . This suggests that the insurer achieves the highest profit when the policyholder places more weight on extreme loss events.

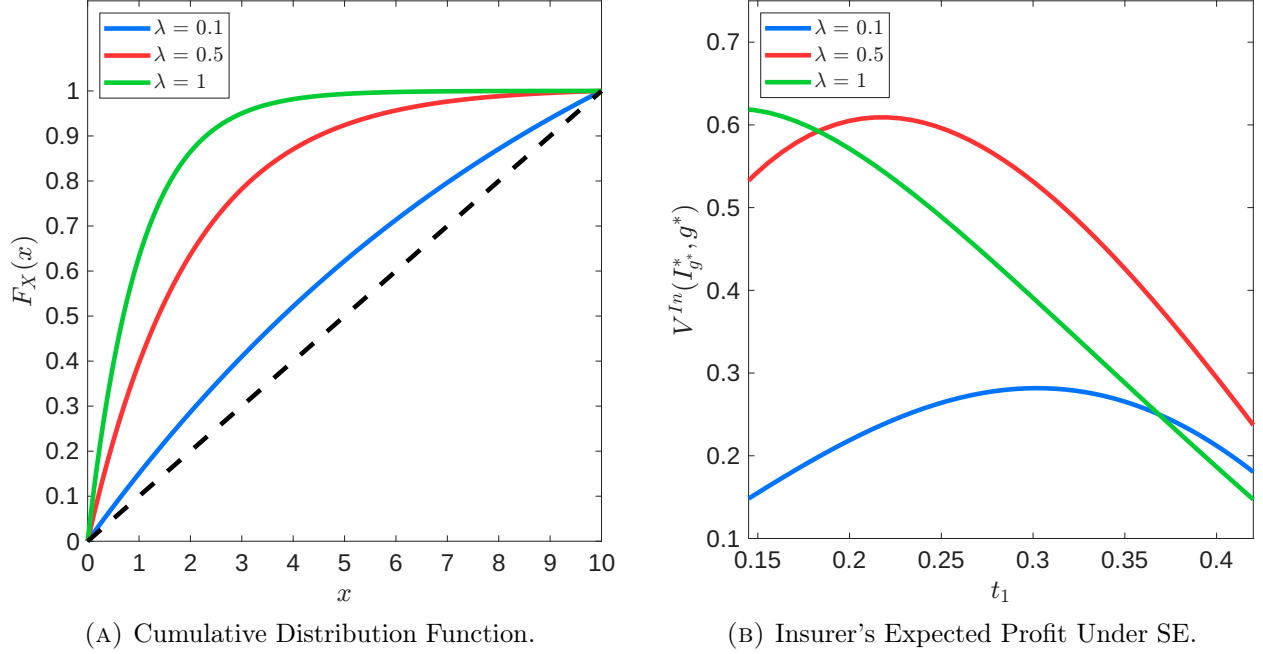


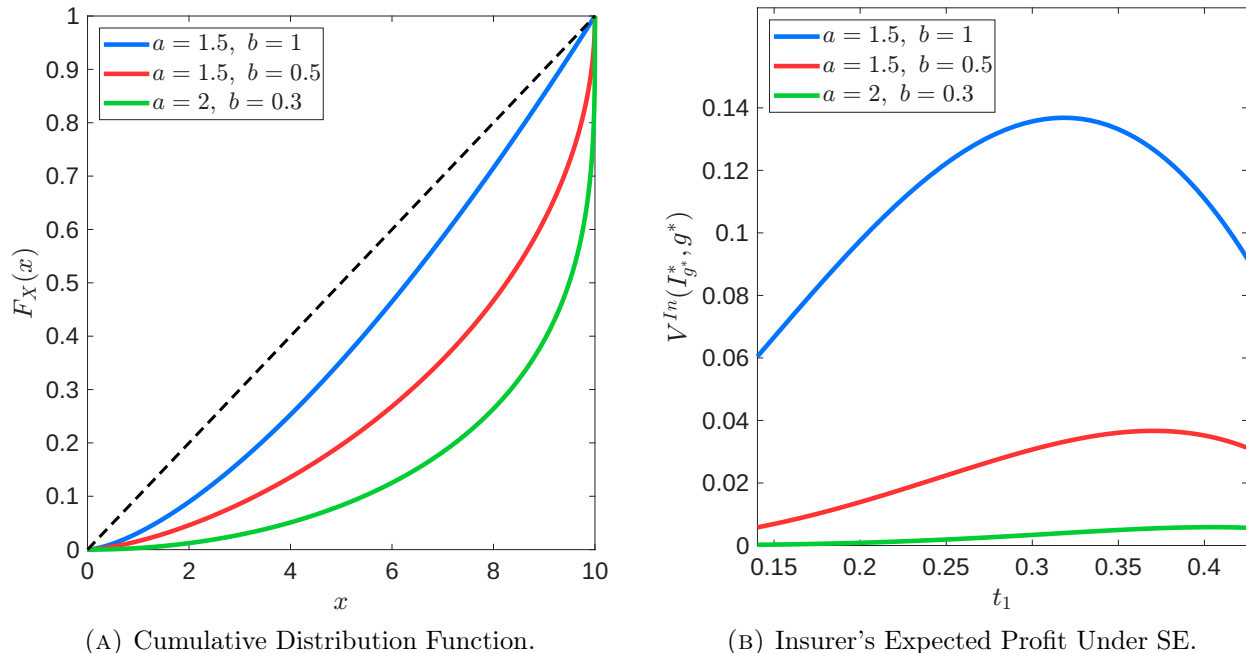
FIGURE 2. The case where X follows a truncated exponential distribution.

Finally, assume that the random loss X follows a Kumaraswamy distribution with CDF given by

$$F_X(x) = 1 - \left(1 - \left(\frac{x}{M}\right)^a\right)^b, \quad \forall x \in [0, M], \quad M = 10, \quad \text{and } a, b > 0.$$

Figure 3a plots the CDFs for different combinations of a and b . The blue curve lies close to, but slightly below, the identity function, indicating that more losses are concentrated on the right compared to the uniform distribution. The red curve is more concave than the blue one, implying that larger losses are more likely. The green curve represents the most extreme right-skewed case among the three.

Figure 3b shows the corresponding insurer's profit under these three scenarios. When $a = 1.5$ and $b = 1$, the insurer's profit does not vary monotonically with t_1 , and it reaches its maximum around $t_1 \approx 0.32$. When $a = 1.5$ and $b = 0.5$, meaning the loss distribution becomes riskier in the sense of first-order stochastic dominance, the insurer's profit decreases for all t_1 . This suggests that when the underlying risk increases, the insurer's profit declines regardless of the policyholder's risk attitude. Moreover, the maximum shifts to the right, reaching about $t_1 \approx 0.37$. When the loss becomes even riskier with $a = 2$ and $b = 0.3$, a similar decreasing pattern is observed, with the maximum shifting further to the right to approximately $t_1 \approx 0.4$.

FIGURE 3. The case where X follows a Kumaraswamy distribution.

6. CONCLUSION

In this paper, we study Stackelberg Equilibria (Bowley optima) in a monopolistic centralized sequential-move insurance market. We consider a risk-neutral, profit-maximizing insurer who sets premia using a distortion premium principle, and a single policyholder who seeks to minimize a distortion risk measure.

We characterize Stackelberg equilibria explicitly, and we show that, in equilibrium, the optimal indemnity function exhibits a layer-type structure, providing full insurance over any loss layer on which the policyholder is more pessimistic than the insurer's pricing functional about tail losses. Equilibrium contracts provide no coverage over loss layers on which the policyholder is less pessimistic than the insurer's pricing functional about tail losses. When the policyholder and the insurer's pricing functional are equally pessimistic about tail losses, the marginal indemnity may take an arbitrary shape, within the global feasibility constraints. Additionally, in equilibrium, the optimal pricing distortion function is determined by the policyholder's degree of (weak) risk aversion, that is, whether the policyholder's distortion function is above or below the identity function, such that prices never exceed the policyholder's marginal willingness to insure tail losses.

When the policyholder is either weakly risk averse (with a distortion function that lies above the identity function) or strongly risk averse (with a concave distortion function), we show that the optimal contract provides full insurance, and the optimal pricing distortion function coincides with that of the policyholder. For policyholders who evaluate risk using a VaR risk measure at a confidence level $\alpha \in (0, 1)$, the optimal coverage includes an upper limit, by providing full insurance for losses below the α -quantile while leaving the upper tail uninsured. For a policyholder who displays strong sensitivity to extreme losses, captured by an inverse-S-shaped distortion function, the optimal indemnity function takes the form of a deductible contract, whereby extreme losses are fully transferred to the insurer. Furthermore, we show that in a Stackelberg equilibrium, both the

insurance coverage and the insurer's expected profit increase with the policyholder's degree of strong or weak risk aversion. A more risk-averse policyholder will receive greater insurance coverage under an equilibrium optimal contract, and is more valuable to the insurer.

Moreover, we examine the Pareto efficiency of equilibrium contracts. Echoing recent work in the literature, we show that equilibrium contracts are Pareto optimal, but they do not induce a welfare gain to the policyholder, which is unsurprising in a monopoly. Conversely, any Pareto-optimal contract that leaves no welfare gain to the policyholder can be obtained as a Stackelberg equilibrium contract.

APPENDIX A. PROOFS OF MAIN RESULTS

A.1. Proof of Lemma 3.4. For a given pricing distortion function g , assume that q_g is optimal for (9), and we aim to show that $I_g(x) = x - q_g(F_X(x))$ is optimal for (6). Consider any feasible solution I to (6), and let $q(t)$ be the quantile function of the associated retention $R(X)$, for all $t \in [0, 1]$. We saw in (8) that

$$\begin{aligned} \rho^{Pol}(I, g) &= \rho^{Pol}(R(X)) + \Pi_g(I(X)) \\ &= \int_0^1 q'(t) [T(1-t) - g(1-t)] dt + \int_0^1 (F_X^{-1})'(t) g(1-t) dt. \end{aligned}$$

Since q_g is optimal for the problem given in (9), then the following inequality holds:

$$\rho^{Pol}(I, g) \geq \int_0^1 q_g'(t) [T(1-t) - g(1-t)] dt + \int_0^1 (F_X^{-1})'(t) g(1-t) dt,$$

where, by Remark 3.3,

$$q_g(t) = F_X^{-1}(t) - F_{I_g(X)}^{-1}(t), \quad \forall t \in [0, 1].$$

Hence,

$$\begin{aligned} \rho^{Pol}(I, g) &\geq \int_0^1 [F_X^{-1}(t) - F_{I_g(X)}^{-1}(t)]' [T(1-t) - g(1-t)] dt + \int_0^1 (F_X^{-1})'(t) g(1-t) dt \\ &= \int_0^1 (F_X^{-1})'(t) T(1-t) dt - \int_0^1 (F_{I_g(X)}^{-1})'(t) T(1-t) dt + \pi_g(I_g(X)) \\ &= \int (X - I_g(X) + \Pi_g(I_g(X))) dT \circ \mathbb{P} = \rho^{Pol}(I_g, g), \end{aligned}$$

which implies that the indemnity function $I_g(x) = x - q_g(F_X(x))$ is optimal for the policyholder's problem (6).

Conversely, we assume that the indemnity function I_g , given by $I_g(x) = x - q_g(F_X(x))$ is optimal for the policyholder's problem (6), and we aim to show that the quantile q_g is optimal for (9). Consider any feasible solution q to (9) and let $I(x) = x - q(F_X(x))$. We have that:

$$\begin{aligned} \rho^{Pol}(q, g) &= \int_0^1 q'(t) [T(1-t) - g(1-t)] dt + \int_0^1 (F_X^{-1})'(t) g(1-t) dt \\ &= \int (X - I(X) + \Pi_g(I(X))) dT \circ \mathbb{P}. \end{aligned}$$

Since I_g is optimal for (6), then the following inequality holds:

$$\rho^{Pol}(q, g) \geq \int (X - I_g(X) + \Pi_g(I_g(X))) dT \circ \mathbb{P}$$

$$= \int_0^1 q'_g(t) [T(1-t) - g(1-t)] dt + \int_0^1 (F_X^{-1})'(t) g(1-t) dt = \rho^{Pol}(q_g, g),$$

which implies the optimality of q_g for (9). $\square$

A.2. Proof of Lemma 3.5. First, note that

$$\begin{aligned} \rho^{Pol}(q, g) &= \Pi_g(q) + \int_0^1 q'(t) T(1-t) dt \\ &= \int_0^1 ((F_X^{-1})'(t) - q'(t)) g(1-t) dt + \int_0^1 q'(t) T(1-t) dt \\ &= \int_0^1 (F_X^{-1})'(t) g(1-t) dt + \int_0^1 q'(t) (T(1-t) - g(1-t)) dt. \end{aligned}$$

The first term in the final line is independent of q , so minimizing the risk measure $\rho^{Pol}(q, g)$ over $q \in \mathcal{Q}_L$ is equivalent to solving:

$$\min_{q \in \mathcal{Q}_L} \int_0^1 q'(t) (T(1-t) - g(1-t)) dt.$$

By the Marginal Indemnity Function Approach given in Assa (2015), the optimal quantile function q_g must satisfy:

$$(q_g)'(t) = \begin{cases} 0, & g(1-t) < T(1-t), \\ \in [0, (F_X^{-1})'(t)], & g(1-t) = T(1-t), \\ (F_X^{-1})'(t), & g(1-t) > T(1-t). \end{cases} \quad (18)$$

$\square$

A.3. Proof of Theorem 3.9. First, note that the insurer's profit can be expressed as:

$$\begin{aligned} V^{In}(q_g, g) &= \int_0^1 [(F_X^{-1})'(t) - q'_g(t)] g(1-t) dt - \int_0^1 [F_X^{-1}(t) - q_g(t)] dt \\ &= \int_0^1 [(F_X^{-1})'(t) - q'_g(t)] g(1-t) dt - \int_0^1 [(F_X^{-1})'(t) - q'_g(t)] (1-t) dt \\ &= \int_0^1 [(F_X^{-1})'(t) - q'_g(t)] h(t) dt, \end{aligned} \quad (19)$$

where $h(t) := t - 1 + g(1-t)$, for all $t \in [0, 1]$. Consider now the following three sets:

$$\mathcal{A}_1 = \{t \in [0, 1] : h(t) < t - \tilde{T}(t)\}, \mathcal{A}_2 = \{t \in [0, 1] : h(t) = t - \tilde{T}(t)\}, \text{ and } \mathcal{A}_3 = \{t \in [0, 1] : h(t) > t - \tilde{T}(t)\}.$$

where $\tilde{T}$ is the conjugate of the distortion function T . Then it follows from Lemma 3.5 that

$$(q_h)'(t) = \begin{cases} 0, & t \in \mathcal{A}_1, \\ \phi_h(t) \in [0, (F_X^{-1})'(t)], & t \in \mathcal{A}_2, \\ (F_X^{-1})'(t), & t \in \mathcal{A}_3. \end{cases} \quad (20)$$

The insurer's profit in (19) can then be rewritten as follows:

$$V^{In}(q_h, h) = \int_0^1 (F_X^{-1})'(t) h(t) \mathbb{1}_{\mathcal{A}_1}(t) dt + \int_0^1 [(F_X^{-1})'(t) - \phi_h(t)] h(t) \mathbb{1}_{\mathcal{A}_2}(t) dt. \quad (21)$$

Note that on $\mathcal{A}_2$, the optimal retention function allows for some flexibility as long as $q_h \in \mathcal{Q}_L$. The arbitrary choice of $\phi_h(t)$ does not affect the policyholder's risk exposure level, but it does impact the insurer's profit. To maximize the insurer's profit, we must take a further step to determine the value of ϕ_h when $t \in \mathcal{A}_2$. This is achieved by analyzing the profit over a finer partition $\{\mathcal{B}_i\}_{i=1}^3$ such that:

$$\mathcal{B}_1 = \{t \in [0, 1] : \tilde{T}(t) < t\}, \quad \mathcal{B}_2 = \{t \in [0, 1] : \tilde{T}(t) = t\}, \quad \text{and} \quad \mathcal{B}_3 = \{t \in [0, 1] : \tilde{T}(t) > t\}.$$

The insurer's profit in (21) becomes:

$$\begin{aligned} V^{In}(q_h, h) &= \int_0^1 (F_X^{-1})'(t) h(t) \mathbb{1}_{\mathcal{A}_1}(t) dt + \sum_{i=1}^3 \int_0^1 [(F_X^{-1})'(t) - \phi_h(t)] h(t) \mathbb{1}_{\mathcal{A}_2}(t) \cdot \mathbb{1}_{\mathcal{B}_i}(t) dt \\ &= \int_0^1 (F_X^{-1})'(t) h(t) \mathbb{1}_{\mathcal{A}_1}(t) dt + \sum_{i=1}^3 \int_0^1 [(F_X^{-1})'(t) - \phi_h(t)] h(t) \mathbb{1}_{\mathcal{A}_2 \cap \mathcal{B}_i}(t) dt. \end{aligned} \quad (22)$$

We consider the following three cases:

- (1) On $\mathcal{A}_2 \cap \mathcal{B}_1$, $h(t) = t - \tilde{T}(t) > 0$. Thus, the insurer's profit in (22) is decreasing in $\phi_h(t)$, and it is optimal to set $\phi_h(t) = 0$.
- (2) On $\mathcal{A}_2 \cap \mathcal{B}_2$, $h(t) = t - \tilde{T}(t) = 0$. The profit contribution is always zero regardless of the value of $\phi_h(t)$. Thus, $\phi_h(t)$ can take any value in $[0, (F_X^{-1})'(t)]$ without affecting the insurer's profit.
- (3) On $\mathcal{A}_2 \cap \mathcal{B}_3$, $h(t) = t - \tilde{T}(t) < 0$. The insurer's profit is increasing in $\phi_h(t)$. Thus, it is optimal to set $\phi_h(t) = (F_X^{-1})'(t)$.

Hence, for a given h , the marginal quantile of the optimal retention can be written as:

$$(q_h^*)'(t) = \begin{cases} 0, & t \in \mathcal{A}_2 \cap \mathcal{B}_1, \\ \phi_h(t) \in [0, (F_X^{-1})'(t)], & t \in \mathcal{A}_2 \cap \mathcal{B}_2, \\ (F_X^{-1})'(t), & t \in \mathcal{A}_2 \cap \mathcal{B}_3. \end{cases} \quad (23)$$

It then follows from (20) and (23) that:

$$(q_h^*)'(t) = \begin{cases} 0, & t \in \mathcal{A}_1 \cup (\mathcal{A}_2 \cap \mathcal{B}_1), \\ \phi_h(t) \in [0, (F_X^{-1})'(t)], & t \in \mathcal{A}_2 \cap \mathcal{B}_2, \\ (F_X^{-1})'(t), & t \in \mathcal{A}_3 \cup (\mathcal{A}_2 \cap \mathcal{B}_3). \end{cases} \quad (24)$$

Using (24) and the fact that $h(t) = 0$ on $\mathcal{A}_2 \cap \mathcal{B}_2$, the insurer's profit in (22) reduces to:

$$\begin{aligned} V^{In}(q_h^*, h) &= \int_0^1 (F_X^{-1})'(t) h(t) [\mathbb{1}_{\mathcal{A}_1}(t) + \mathbb{1}_{\mathcal{A}_2 \cap \mathcal{B}_1}(t)] dt \\ &= \int_0^1 (F_X^{-1})'(t) h(t) \mathbb{1}_{\mathcal{A}_1 \cup (\mathcal{A}_2 \cap \mathcal{B}_1)}(t) dt, \text{ since } \mathcal{A}_1 \text{ and } \mathcal{A}_2 \text{ are disjoint.} \end{aligned}$$

Moreover,

$$\mathcal{A}_1 \cup (\mathcal{A}_2 \cap \mathcal{B}_1) = (\mathcal{A}_1 \cup \mathcal{A}_2) \cap (\mathcal{A}_1 \cup \mathcal{B}_1).$$

Since $\{\mathcal{B}_i\}_{i=1}^3$ forms a partition of $[0, 1]$, we have:

$$\mathcal{A}_1 \cup \mathcal{B}_1 = \mathcal{B}_1 \cup (\mathcal{A}_1 \cap (\mathcal{B}_2 \cup \mathcal{B}_3)).$$

Combining the above identities yields:

$$\mathcal{A}_1 \cup (\mathcal{A}_2 \cap \mathcal{B}_1) = [(\mathcal{A}_1 \cup \mathcal{A}_2) \cap \mathcal{B}_1] \cup [\mathcal{A}_1 \cap (\mathcal{B}_2 \cup \mathcal{B}_3)],$$

which is a union of disjoint sets. Therefore,

$$V^{In}(q_h^*, h) = \int_0^1 (F_X^{-1})'(t) h(t) [\mathbf{1}_{(\mathcal{A}_1 \cup \mathcal{A}_2) \cap \mathcal{B}_1}(t) + \mathbf{1}_{\mathcal{A}_1 \cap (\mathcal{B}_2 \cup \mathcal{B}_3)}(t)] dt.$$

The insurer's optimization problem given in (12) thus reduces to the following:

$$\max_h \left\{ \int_0^1 (F_X^{-1})'(t) h(t) [\mathbb{1}_{\{h(t) \leq t - \tilde{T}(t)\}} \mathbb{1}_{\{\tilde{T}(t) < t\}} + \mathbb{1}_{\{h(t) < t - \tilde{T}(t)\}} \mathbb{1}_{\{\tilde{T}(t) \geq t\}}] dt \right\}. \quad (25)$$

Since the integrand depends on h only through the pointwise value $h(t)$, the above problem (25) can be solved by maximizing the integrand pointwise. Fix $t_0 \in (0, 1)$ and consider the auxiliary maximization problem:

$$\max_{y \geq 0} m(y; t_0),$$

where the auxiliary function $m(y; t_0)$ is given by

$$m(y; t_0) := y [\mathbb{1}_{\{y \leq t_0 - \tilde{T}(t_0)\}} \mathbb{1}_{\{\tilde{T}(t_0) < t_0\}} + \mathbb{1}_{\{y < t_0 - \tilde{T}(t_0)\}} \mathbb{1}_{\{\tilde{T}(t_0) \geq t_0\}}].$$

For a fixed t_0 , the function $m(y; t_0)$ is piecewise linear in y . Let $y_{t_0} \in \arg \max_y m(y; t_0)$. The maximum value of $m(y; t_0)$ is given by

$$m(y_{t_0}; t_0) = \begin{cases} t_0 - \tilde{T}(t_0), & \tilde{T}(t_0) < t_0, \\ 0, & \tilde{T}(t_0) \geq t_0, \end{cases}$$

and any maximizer y_{t_0} satisfies:

$$\begin{cases} y_{t_0} = t_0 - \tilde{T}(t_0), & \text{if } \tilde{T}(t_0) < t_0, \\ y_{t_0} \geq t_0 - \tilde{T}(t_0), & \text{if } \tilde{T}(t_0) \geq t_0. \end{cases} \quad (26)$$

for any $t_0 \in (0, 1)$. We now proceed to the characterization of optimal solutions to (25) through the following Lemma.

Lemma A.1. *A function h^* is optimal to (25) if and only if for every $t_0 \in (0, 1)$, $h^*(t_0) = y_{t_0}$, where y_{t_0} satisfies (26).*

Proof. Assume that h^* is a solution to (25). We show that $h^*(t_0) = y_{t_0}$, where y_{t_0} satisfies (26). We consider the following two cases.

- (1) If $\tilde{T}(t_0) < t_0$, we aim to show that $h^*(t_0) = t_0 - \tilde{T}(t_0)$ for any $t_0 \in (0, 1)$. Suppose for the sake of contradiction that

$$h^*(t_0) \neq t_0 - \tilde{T}(t_0).$$

Since $\tilde{T}(t_0) < t_0$ and $\tilde{T}$ is continuous (being differentiable), we can find an arbitrary small $\epsilon > 0$ such that

$$\tilde{T}(t) < t, \quad \text{when } t \in (t_0 - \epsilon, t_0 + \epsilon).$$

If $h^*(t_0) < t_0 - \tilde{T}(t_0)$, we have $m(h^*(t_0); t_0) = h^*(t_0)$. Then there exists a function $\tilde{h}$ such that:

$$\tilde{h}(t) = \begin{cases} y_t > h^*(t), & t \in (t_0 - \epsilon, t_0 + \epsilon), \\ h^*(t), & t \notin (t_0 - \epsilon, t_0 + \epsilon), \end{cases}$$

where y_t satisfies (26), which implies the following:

$$\begin{aligned} V^{In}(q_h^*, \tilde{h}) - V^{In}(q_{h^*}^*, h^*) &= \int_{t_0 - \epsilon}^{t_0 + \epsilon} (F_X^{-1})'(t) [m(y_t; t) - m(h^*(t); t)] dt \\ &= \int_{t_0 - \epsilon}^{t_0 + \epsilon} (F_X^{-1})'(t) [y_t - h^*(t)] dt > 0, \end{aligned}$$

since $y_t > h^*(t)$ when $t \in (t_0 - \epsilon, t_0 + \epsilon)$. Hence, this contradicts the fact that h^* is optimal for problem (25).

If, in contrast, $h^*(t_0) > t_0 - \tilde{T}(t_0)$, then $m(h^*(t_0); t_0) = 0$. With the same ϵ , there exists a function $\tilde{h}$ such that:

$$\tilde{h}(t) = \begin{cases} y_t < h^*(t), & t \in (t_0 - \epsilon, t_0 + \epsilon), \\ h^*(t), & t \notin (t_0 - \epsilon, t_0 + \epsilon). \end{cases}$$

Hence,

$$\begin{aligned} V^{In}(q_h^*, \tilde{h}) - V^{In}(q_{h^*}^*, h^*) &= \int_{t_0 - \epsilon}^{t_0 + \epsilon} (F_X^{-1})'(t) [m(y_t; t) - m(h^*(t); t)] dt \\ &= \int_{t_0 - \epsilon}^{t_0 + \epsilon} (F_X^{-1})'(t) (y_t - 0) dt > 0, \end{aligned}$$

which also contradicts the optimality of h^* for (25). Therefore, $h^*(t_0) = t_0 - \tilde{T}(t_0)$, for any $t_0 \in (0, 1)$, when $\tilde{T}(t_0) < t_0$.

- (2) If $\tilde{T}(t_0) \geq t_0$, we aim to show that $h^*(t_0) \geq t_0 - \tilde{T}(t_0)$, for any $t_0 \in (0, 1)$. Suppose for the sake of contradiction that

$$h^*(t_0) < t_0 - \tilde{T}(t_0).$$

Then

$$m(h^*(t_0); t_0) = h^*(t_0) < t_0 - \tilde{T}(t_0) \leq 0.$$

There exists an arbitrary small $\epsilon > 0$ such that $\tilde{T}(t) \geq t$ on $(t_0 - \epsilon, t_0]$ or on $[t_0, t_0 + \epsilon)$. First, assume that $\tilde{T}(t) \geq t$ on $(t_0 - \epsilon, t_0]$. Then there exists a function $\tilde{h}$ such that:

$$\tilde{h}(t) = \begin{cases} y_t > h^*(t), & t \in (t_0 - \epsilon, t_0], \\ h^*(t), & t \notin (t_0 - \epsilon, t_0]. \end{cases}$$

Therefore,

$$\begin{aligned} V^{In}(q_h^*, \tilde{h}) - V^{In}(q_{h^*}^*, h^*) &= \int_{t_0 - \epsilon}^{t_0} (F_X^{-1})'(t) [m(y_t; t) - m(h^*(t); t)] dt \\ &= \int_{t_0 - \epsilon}^{t_0} (F_X^{-1})'(t) [0 - m(h^*(t); t)] dt > 0, \end{aligned}$$

since $m(h^*(t); t) < 0$ when $t \in (t_0 - \epsilon, t_0]$, which leads to a contradiction. Hence,

$$h^*(t_0) \geq t_0 - \tilde{T}(t_0).$$

Moreover, we can derive a similar result if $\tilde{T}(t) \geq t$ on $[t_0, t_0 + \epsilon)$. In sum, for any $t_0 \in [0, 1]$, we obtain:

$$\begin{cases} h^*(t_0) = t_0 - \tilde{T}(t_0), & \text{if } \tilde{T}(t_0) < t_0, \\ h^*(t_0) \geq t_0 - \tilde{T}(t_0), & \text{if } \tilde{T}(t_0) \geq t_0. \end{cases} \quad (27)$$

Conversely, assume that h^* satisfies (27). We show that h^* is optimal for (25). We first note that the insurer's expected profit under $(q_{h^*}^*, h^*)$ is given by

$$V^{In}(q_{h^*}^*, h^*) = \int_0^1 (F_X^{-1})'(t) (t - \tilde{T}(t)) \mathbb{1}_{\{\tilde{T}(t) < t\}} dt.$$

For any other feasible solution h , we compare the insurer's expected profit under h^* and h . Specifically, from the structure of (25), we have:

$$\begin{aligned} & V^{In}(q_{h^*}^*, h^*) - V^{In}(q_h^*, h) \\ &= \int_0^1 (F_X^{-1})'(t) (t - \tilde{T}(t)) \mathbb{1}_{\{\tilde{T}(t) < t\}} dt \\ &\quad - \int_0^1 (F_X^{-1})'(t) h(t) \left[\mathbb{1}_{\{h(t) \leq t - \tilde{T}(t)\}} \mathbb{1}_{\{\tilde{T}(t) < t\}}(t) + \mathbb{1}_{\{h(t) < t - \tilde{T}(t)\}} \mathbb{1}_{\{\tilde{T}(t) = t\}}(t) + \mathbb{1}_{\{h(t) < t - \tilde{T}(t)\}} \mathbb{1}_{\{\tilde{T}(t) > t\}}(t) \right] dt. \end{aligned}$$

That is,

$$\begin{aligned} V^{In}(q_{h^*}^*, h^*) - V^{In}(q_h^*, h) &= \int_0^1 (F_X^{-1})'(t) [t - \tilde{T}(t) - h(t) \mathbb{1}_{\{h(t) \leq t - \tilde{T}(t)\}}] \mathbb{1}_{\{\tilde{T}(t) < t\}} dt \\ &\quad - \int_0^1 (F_X^{-1})'(t) h(t) [\mathbb{1}_{\{h(t) < t - \tilde{T}(t)\}} \mathbb{1}_{\{\tilde{T}(t) = t\}} + \mathbb{1}_{\{h(t) < t - \tilde{T}(t)\}} \mathbb{1}_{\{\tilde{T}(t) > t\}}] dt. \end{aligned} \quad (28)$$

Looking at the first term of (28), we know that

$$t - \tilde{T}(t) - h(t) \mathbb{1}_{\{h(t) \leq t - \tilde{T}(t)\}} = \begin{cases} t - \tilde{T}(t), & h(t) > t - \tilde{T}(t) \\ t - \tilde{T}(t) - h(t), & h(t) \leq t - \tilde{T}(t), \end{cases}$$

which is always nonnegative. For the second term,

$$-h(t) \mathbb{1}_{\{h(t) < t - \tilde{T}(t)\}} \begin{cases} = h(t) \mathbb{1}_{\{h(t) < 0\}} \geq 0, & \text{if } \tilde{T}(t) = t, \\ \geq 0, & \text{if } \tilde{T}(t) > t. \end{cases}$$

Thus, both terms of (28) are nonnegative, for all $t \in [0, 1]$, implying that the integrand is pointwise nonnegative. Therefore, the difference in profit is nonnegative, i.e.

$$V^{In}(q_{h^*}^*, h^*) \geq V^{In}(q_h^*, h).$$

Hence, h^* maximizes the insurer's profit functional, thereby establishing sufficiency. $\square$

Moreover, when h^* satisfies (27), we can characterize the structure of the critical sets $\mathcal{A}_1 \cup (\mathcal{A}_2 \cap \mathcal{B}_1)$, $\mathcal{A}_2 \cap \mathcal{B}_2$, and $\mathcal{A}_3 \cup (\mathcal{A}_2 \cap \mathcal{B}_3)$ that define $q_{h^*}^*$ in (24). First, if h^* satisfies (27), we can clearly see that $\mathcal{A}_1 = \emptyset$. Hence,

$$\mathcal{A}_1 \cup (\mathcal{A}_2 \cap \mathcal{B}_1) = \{t \in [0, 1] : t - \tilde{T}(t) > 0\}.$$

Moreover,

$$\mathcal{A}_2 \cap \mathcal{B}_2 = \{t \in [0, 1] : h^*(t) = t - \tilde{T}(t) = 0\},$$

and

$$\mathcal{A}_3 \cup (\mathcal{A}_2 \cap \mathcal{B}_3) = \{t \in [0, 1] : h^*(t) > t - \tilde{T}(t)\} \cup \{t \in [0, 1] : h^*(t) = t - \tilde{T}(t) < 0\}.$$

Additionally, since $\{\mathcal{B}_i\}_{i=1}^3$ forms a partition over $(0, 1)$, it follows that

$$\begin{aligned} \mathcal{A}_3 &= (\mathcal{A}_3 \cap \mathcal{B}_1) \cup (\mathcal{A}_3 \cap \mathcal{B}_2) \cup (\mathcal{A}_3 \cap \mathcal{B}_3) \\ &= \{t \in [0, 1] : h^*(t) > t - \tilde{T}(t), t > \tilde{T}(t)\} \\ &\quad \cup \{t \in [0, 1] : h^*(t) > t - \tilde{T}(t) = 0\} \\ &\quad \cup \{t \in [0, 1] : h(t)^* > t - \tilde{T}(t), \tilde{T}(t) > t\}. \end{aligned}$$

The first intersection must be empty since h^* satisfies (27):

$$\mathcal{A}_3 \cap \mathcal{B}_1 = \emptyset.$$

Hence, the union of $\mathcal{A}_3$ with $\mathcal{A}_2 \cap \mathcal{B}_3$ reduces to the following:

$$\mathcal{A}_3 \cup (\mathcal{A}_2 \cap \mathcal{B}_3) = \{t \in [0, 1] : h^*(t) > t - \tilde{T}(t) = 0\} \cup \{t \in [0, 1] : \tilde{T}(t) > t\}.$$

Thus, the function $q_{h^*}^*$ given in (24) can be written as follows:

$$(q_{h^*}^*)'(t) = \begin{cases} 0, & \tilde{T}(t) < t, \\ \phi_h(t) \in [0, (F_X^{-1})'(t)], & t \in \{z; h^*(z) = z - \tilde{T}(z) = 0\} \\ (F_X^{-1})'(t), & t \in \{z; h^*(z) > z - \tilde{T}(z) = 0\} \cup \{z; \tilde{T}(z) > z\}. \end{cases}$$

Since $\tilde{g}^*(t) = t - h^*(t)$, the optimal h^* leads to the following optimal pricing distortion function $\tilde{g}^*$:

$$\tilde{g}^*(t) = \begin{cases} \tilde{T}(t), & \tilde{T}(t) < t, \\ \in [0, \tilde{T}(t)], & \tilde{T}(t) \geq t. \end{cases} \quad (29)$$

Consequently,

$$\tilde{g}^*(t) = \begin{cases} \tilde{T}(t), & \tilde{T}(t) < t, \\ \in [\sup\{z < t; \tilde{g}^*(z)\}, \tilde{T}(t)], & \tilde{T}(t) \geq t. \end{cases} \quad (30)$$

We then obtain the optimal retention quantile q^* as a function of the distortion premium function g^* , as follows:

$$(q_{g^*}^*)'(t) = \begin{cases} 0, & \tilde{T}(t) < t, \\ \phi_{g^*}(t) \in [0, (F_X^{-1})'(t)], & t \in \{z; \tilde{g}^*(z) = \tilde{T}(z) = z\}, \\ (F_X^{-1})'(t), & t \in \{z; \tilde{g}^*(z) < \tilde{T}(z) = z\} \cup \{z; \tilde{T}(z) > z\}. \end{cases}$$

□

A.4. Proof of Theorem 3.11. Consider two policyholders whose respective distortion functions T_1 and T_2 satisfy $T_1(t) \leq T_2(t)$ for all $t \in [0, 1]$. Then

$$T_1(\mathbb{P}[X > y]) \leq T_2(\mathbb{P}[X > y]), \quad \forall y \in [0, M].$$

Let $(I_{g_1}^*, g_1^*)$ and $(I_{g_2}^*, g_2^*)$ denote the corresponding Stackelberg equilibria, where κ_1^* and κ_2^* denote the marginal indemnity functions of the two policyholders, respectively, and satisfy Corollary 3.10. Hence, we obtain

$$\kappa_1^*(y) \leq \kappa_2^*(y), \text{ for almost every } y \in [0, M].$$

The above inequality gives

$$I_{g_1}^*(x) \leq I_{g_2}^*(x), \quad \forall x \in [0, M].$$

The insurer's expected profits for the two policyholders are given by:

$$V^{In}(I_{g_1}^*, g_1^*) = \int_0^1 (F_X^{-1})'(t) [t - \tilde{T}_1(t)] \mathbb{1}_{\{\tilde{T}_1(t) < t\}} dt \quad \text{and} \quad V^{In}(I_{g_2}^*, g_2^*) = \int_0^1 (F_X^{-1})'(t) [t - \tilde{T}_2(t)] \mathbb{1}_{\{\tilde{T}_2(t) < t\}} dt,$$

where $\tilde{T}_1$ and $\tilde{T}_2$ denote the conjugates of T_1 and T_2 , respectively. Since $T_1(t) \leq T_2(t)$ for all $t \in [0, 1]$, it follows that $\tilde{T}_1(t) \geq \tilde{T}_2(t)$, for all $t \in [0, 1]$. Hence,

$$t - \tilde{T}_1(t) \leq t - \tilde{T}_2(t), \quad \forall t \in [0, 1],$$

and,

$$\{\tilde{T}_1(t) < t\} \subseteq \{\tilde{T}_2(t) < t\}.$$

Therefore,

$$\begin{aligned} V^{In}(I_{g_1}^*, g_1^*) &= \int_0^1 (F_X^{-1})'(t) [t - \tilde{T}_1(t)] \mathbb{1}_{\{\tilde{T}_1(t) < t\}} dt \\ &\leq \int_0^1 (F_X^{-1})'(t) [t - \tilde{T}_2(t)] \mathbb{1}_{\{\tilde{T}_2(t) < t\}} dt = V^{In}(I_{g_2}^*, g_2^*). \end{aligned}$$

Hence, the second policyholder is more profitable for the insurer than the first. $\square$

A.5. Proof of Proposition 4.1. Let $(I^*, \pi^*) \in \mathcal{I}_L \times \mathbb{R}$ be optimal for the problem given in (17), and assume for the sake of contradiction that (I^*, π^*) is not Pareto optimal. Then there exists a contract $(I, \pi) \in \mathcal{I}_L \times \mathbb{R}$ such that

$$\rho^{Pol}(I, \pi) \leq \rho^{Pol}(I^*, \pi^*) \quad \text{and} \quad V^{In}(I, \pi) \geq V^{In}(I^*, \pi^*),$$

with at least one strict inequality. Consequently,

$$\rho^{Pol}(I, \pi) - V^{In}(I, \pi) < \rho^{Pol}(I^*, \pi^*) - V^{In}(I^*, \pi^*),$$

which contradicts the optimality of (I^*, π^*) for the problem given in (17). Hence, the contract (I^*, π^*) is Pareto optimal.

Conversely, suppose that the contract (I^*, π^*) is Pareto optimal, and assume for the sake of contradiction that (I^*, π^*) is not optimal for (17). Then there exists a contract $(\tilde{I}, \tilde{\pi}) \in \mathcal{I}_L \times \mathbb{R}$ such that

$$\rho^{Pol}(\tilde{I}, \tilde{\pi}) - V^{In}(\tilde{I}, \tilde{\pi}) < \rho^{Pol}(I^*, \pi^*) - V^{In}(I^*, \pi^*).$$

Let $\hat{\pi} := \tilde{\pi} + \rho^{Pol}(I^*, \pi^*) - \rho^{Pol}(\tilde{I}, \tilde{\pi})$. Then by translation invariance of ρ^{Pol} , we have:

$$\rho^{Pol}(\tilde{I}, \hat{\pi}) = \rho^{Pol}(X - \tilde{I}(X)) + \hat{\pi} = \rho^{Pol}(I^*, \pi^*).$$

Moreover,

$$\begin{aligned} V^{In}(\tilde{I}, \hat{\pi}) - V^{In}(I^*, \pi^*) &= \hat{\pi} - \mathbb{E}[\tilde{I}(X)] - V^{In}(I^*, \pi^*) \\ &= (\tilde{\pi} + \rho^{Pol}(I^*, \pi^*) - \rho^{Pol}(\tilde{I}, \tilde{\pi})) - \mathbb{E}[\tilde{I}(X)] - V^{In}(I^*, \pi^*) \\ &= \rho^{Pol}(I^*, \pi^*) - \rho^{Pol}(\tilde{I}, \tilde{\pi}) + V^{In}(\tilde{I}, \tilde{\pi}) - V^{In}(I^*, \pi^*) > 0. \end{aligned}$$

Thus, the contract $(\tilde{I}, \hat{\pi})$ satisfies

$$\rho^{Pol}(\tilde{I}, \hat{\pi}) = \rho^{Pol}(I^*, \pi^*) \quad \text{and} \quad V^{In}(\tilde{I}, \hat{\pi}) > V^{In}(I^*, \pi^*),$$

which contradicts the Pareto optimality of (I^*, π^*) . Hence, (I^*, π^*) is optimal for (17). $\square$

A.6. Proof of Proposition 4.2. Consider a Stackelberg equilibrium $(I_{g^*}^*, g^*)$ inducing a contract $(I_{g^*}^*, \pi_{g^*}^*)$, where $\pi_{g^*}^* := \Pi_{g^*}(I_{g^*}^*(X)) = \int I_{g^*}^* dg^* \circ \mathbb{P}$. By Proposition 4.1, to show that $(I_{g^*}^*, \pi_{g^*}^*)$ is Pareto optimal, it suffices to show that it is optimal for Problem (17). By translation invariance of the distortion risk measure, a contract (I^*, π^*) is Pareto optimal if and only if the indemnity function I^* solves

$$\min_{I \in \mathcal{I}_L} \left\{ \rho^{Pol}(X - I(X)) + \mathbb{E}[I(X)] \right\}.$$

Equivalently, using retention functions, the above problem becomes:

$$\min_{R \in \mathcal{I}_L} \left\{ \rho^{Pol}(R(X)) + \mathbb{E}[X - R(X)] \right\}.$$

Using $q(t) := F_{R(X)}^{-1}(t)$ for a.e. $t \in [0, 1]$, the above problem is equivalent to the following:

$$\min_{q \in \mathcal{Q}_L} \left\{ \int_0^1 q'(t) [T(1-t) - (1-t)] dt + \mathbb{E}[X] \right\}.$$

Since $(I_{g^*}^*, g^*)$ is a Stackelberg equilibrium, the optimal quantile $q_{g^*}^*$ characterized in Theorem 3.9 solves the above minimization problem. Hence, the induced contract $(I_{g^*}^*, \pi_{g^*}^*)$ is Pareto optimal. We now show that $(I_{g^*}^*, \pi_{g^*}^*)$ is individually rational but leaves the policyholder indifferent. First, note that the insurer's expected profit under $(I_{g^*}^*, \pi_{g^*}^*)$ satisfies

$$V^{In}(I_{g^*}^*, \pi_{g^*}^*) = \int_0^1 (F_X^{-1})'(t) [t - \tilde{T}(t)] \mathbb{1}_{\{\tilde{T}(t) < t\}} dt \geq 0 = V^{In}(0, 0).$$

Moreover, for the policyholder, we have

$$\begin{aligned} \rho^{Pol}(I_{g^*}^*, \pi_{g^*}^*) - \rho^{Pol}(0, 0) &= \int_0^1 (F_X^{-1})'(t) g^*(1-t) dt + \int_0^1 (q_{g^*}^*)'(t) [T(1-t) - g^*(1-t)] dt \\ &\quad - \int_0^1 (F_X^{-1})'(t) T(1-t) dt \\ &= \int_0^1 [(q_{g^*}^*)'(t) - (F_X^{-1})'(t)] [T(1-t) - g^*(1-t)] dt. \end{aligned}$$

Consider the following three cases:

- (1) If $\tilde{T}(t) < t$, then by optimality, $\tilde{g}^*(t) = \tilde{T}(t)$, or equivalently $g^*(1-t) = T(1-t)$. Hence, $[(q_{g^*}^*)'(t) - (F_X^{-1})'(t)] [T(1-t) - g^*(1-t)] = 0$.

(2) If $\tilde{T}(t) = t$ and $\tilde{g}^*(t) < \tilde{T}(t)$, then $(q_{g^*}^*)'(t) = (F_X^{-1})'(t)$. If $\tilde{T}(t) = t$ and $\tilde{g}^*(t) = \tilde{T}(t)$, then $g^*(1-t) = T(1-t)$. In both cases, $[(q_{g^*}^*)'(t) - (F_X^{-1})'(t)] [T(1-t) - g^*(1-t)] = 0$.

(3) If $\tilde{T}(t) > t$, then $(q_{g^*}^*)'(t) = (F_X^{-1})'(t)$, and $[(q_{g^*}^*)'(t) - (F_X^{-1})'(t)] [T(1-t) - g^*(1-t)] = 0$.

Consequently, we finally obtain $\rho^{Pol}(I_{g^*}^*, \pi_{g^*}^*) = \rho^{Pol}(0, 0)$. $\square$

A.7. Proof of Proposition 4.3. Consider a Pareto optimal contract (I^*, π^*) , such that $\rho^{Pol}(I^*, \pi^*) = \rho^{Pol}(0, 0)$. We show that this contract is induced by a Stackelberg equilibrium. First, note that under (I^*, π^*) , the following holds by translation invariance and comonotonic additivity of ρ^{Pol} :

$$\rho^{Pol}(I^*, \pi^*) = \rho^{Pol}(X - I^*(X)) + \pi^* = \rho^{Pol}(X) - \rho^{Pol}(I^*(X)) + \pi^*.$$

Since, $\rho^{Pol}(I^*, \pi^*) = \rho^{Pol}(0, 0) = \rho^{Pol}(X)$, we have that $\pi^* = \rho^{Pol}(I^*(X))$. Moreover, since (I^*, π^*) is Pareto optimal, Proposition 4.1 implies that the quantile function $q^*(t) := F_X^{-1}(t) - F_{I^*}^{-1}(t)$ satisfies

$$(q^*)'(t) = \begin{cases} 0, & T(1-t) > 1-t, \\ \phi(t), & T(1-t) = 1-t, \\ (F_X^{-1})'(t), & T(1-t) < 1-t, \end{cases}$$

for some measurable function ϕ satisfying $\phi(t) \in [0, (F_X^{-1})'(t)]$.

Consider now the pair $(q^*, g_{I^*}^*)$, where the pricing distortion function $g_{I^*}^*$ coincides with the policyholder's distortion function T . That is,

$$g_{I^*}^*(t) := T(t), \quad \forall t \in [0, 1].$$

Thus,

$$\Pi_{g_{I^*}^*}(I^*(X)) = \int I^*(X) dg_{I^*}^* \circ \mathbb{P} = \int I^*(X) dT \circ \mathbb{P} = \rho^{Pol}(I^*(X)) = \pi^*.$$

Since $(q^*, g_{I^*}^*)$ satisfies the optimality conditions of Theorem 3.9, it follows that $(q^*, g_{I^*}^*)$ solves the problem given in (12). Lemma 3.8 then implies that the market mechanism $(I^*, g_{I^*}^*)$ is a Stackelberg equilibrium. Hence, the Pareto optimal insurance contract (I^*, π^*) can be obtained from the Stackelberg equilibrium $(I^*, g_{I^*}^*)$, where $\pi^* := \Pi_{g_{I^*}^*}(I^*(X))$. $\square$

REFERENCES

- ANDRAOS, M., M. GHOSOUB, AND M. B. ZHU (2026): "Subgame Perfect Nash Equilibria in Large Reinsurance Markets," *Insurance: Mathematics and Economics*, 127, 103210.
- ASSA, H. (2015): "On Optimal Reinsurance Policy with Distortion Risk Measures and Premiums," *Insurance: Mathematics and Economics*, 61, 70–75.
- BAI, Y., Z. ZHOU, H. XIAO, AND R. GAO (2021): "A Stackelberg Reinsurance–Investment Game with Asymmetric Information and Delay," *Optimization*, 70, 2131–2168.
- BAI, Y., Z. ZHOU, H. XIAO, R. GAO, AND F. ZHONG (2022): "A Hybrid Stochastic Differential Reinsurance and Investment Game with Bounded Memory," *European Journal of Operational Research*, 296, 717–737.
- BLEICHRODT, H., S. GRANT, AND J. YANG (2023): "Testing Hurwicz Expected Utility," *Econometrica*, 91, 1393–1416.

- BOONEN, T., K. CHEUNG, AND Y. ZHANG (2021): “Bowley Reinsurance with Asymmetric Information on the Insurer’s Risk Preferences,” *Scandinavian Actuarial Journal*, 2021, 623–644.
- BOONEN, T. AND Y. ZHANG (2022): “Bowley Reinsurance with Asymmetric Information: A First-Best Solution,” *Scandinavian Actuarial Journal*, 2022, 532–551.
- BOONEN, T. J. AND M. GHOSOUB (2023): “Bowley vs. Pareto Optima in Reinsurance Contracting,” *European Journal of Operational Research*, 307, 382–391.
- BOWLEY, A. L. (1928): “Bilateral Monopoly,” *Economic Journal*, 38, 651–659.
- CAO, J., D. LI, V. R. YOUNG, AND B. ZOU (2022): “Stackelberg Differential Game for Insurance under Model Ambiguity,” *Insurance: Mathematics and Economics*, 106, 128–145.
- (2023a): “Reinsurance Games with Two Reinsurers: Tree versus Chain,” *European Journal of Operational Research*, 310, 928–941.
- (2023b): “Stackelberg Differential Game for Insurance under Model Ambiguity: General Divergence,” *Scandinavian Actuarial Journal*, 2023, 735–763.
- (2024): “Stackelberg Reinsurance Chain under Model Ambiguity,” *Scandinavian Actuarial Journal*, 2024, 329–360.
- CARLIER, G. AND R.-A. DANA (2003): “Pareto Efficient Insurance Contracts when the Insurer’s Cost Function is Discontinuous,” *Economic Theory*, 21, 871–893.
- (2005): “Rearrangement Inequalities in Non-convex Insurance Models,” *Journal of Mathematical Economics*, 41, 483–503.
- CHAN, F.-Y. AND H. GERBER (1985): “The Reinsurer’s Monopoly and the Bowley Solution,” *ASTIN Bulletin: The Journal of the IAA*, 15, 141–148.
- CHEN, L. AND Y. SHEN (2018): “On a New Paradigm of Optimal Reinsurance: A Stochastic Stackelberg Differential Game Between an Insurer and a Reinsurer,” *ASTIN Bulletin*, 48, 905–960.
- (2019): “Stochastic Stackelberg Differential Reinsurance Games under Time-Inconsistent Mean–Variance Framework,” *Insurance: Mathematics and Economics*, 88, 120–137.
- CHEN, L., Y. SHEN, AND J. SU (2020): “A Continuous-Time Theory of Reinsurance Chains,” *Insurance: Mathematics and Economics*, 95, 129–146.
- CHEUNG, K. C., S. C. P. YAM, AND Y. ZHANG (2019): “Risk-Adjusted Bowley Reinsurance under Distorted Probabilities,” *Insurance: Mathematics and Economics*, 86, 64–72.
- CHEW, S., E. KARNI, AND Z. SAFRA (1987): “Risk Aversion in the Theory of Expected Utility with Rank Dependent Probabilities,” *Journal of Economic Theory*, 42, 370–381.
- FÖLLMER, H. AND A. SCHIED (2016): *Stochastic Finance: An Introduction in Discrete Time – 4th ed.*, Walter de Gruyter.
- GHOSOUB, M. AND X. HE (2021): “Comparative Risk Aversion in RDEU with Applications to Optimal Underwriting of Securities Issuance,” *Insurance Mathematics and Economics*, 101, 6–22.
- GHOSOUB, M., B. LI, AND B. SHI (2025): “Bowley-Optimal Convex-Loaded Premium Principles,” *Insurance: Mathematics and Economics*, 121, 157–180.
- GHOSOUB, M. AND M. B. ZHU (2024): “Stackelberg Equilibria with Multiple Policyholders,” *Insurance: Mathematics and Economics*, 116, 189–201.
- G. TAYLOR (1992): “Risk Exchange I: A Unification of Some Existing Results,” *Scandinavian Actuarial Journal*, 1992, 15–39.
- GU, A., F. G. VIENS, AND Y. SHEN (2020): “Optimal Excess-of-Loss Reinsurance Contract with Ambiguity Aversion in the Principal-Agent Model,” *Scandinavian Actuarial Journal*, 2020, 342–375.
- LI, D. AND V. R. YOUNG (2021): “Bowley Solution of a Mean–Variance Game in Insurance,” *Insurance: Mathematics and Economics*, 98, 35–43.
- (2022): “Stackelberg Differential Game for Reinsurance: Mean-Variance Framework and Random Horizon,” *Insurance: Mathematics and Economics*, 102, 42–55.
- PRELEC, D. (1998): “The Probability Weighting Function,” *Econometrica*, 66, 497–527.

- QUIGGIN, J. (1993): *Generalized Expected Utility Theory - The Rank-Dependent Model*, Kluwer Academic Publishers.
- TVERSKY, A. AND D. KAHNEMAN (1992): “Advances in Prospect Theory: Cumulative Representation of Uncertainty,” *Journal of Risk and Uncertainty*, 5, 297–323.
- WANG, N. AND T. K. SIU (2020): “Robust Reinsurance Contracts with Risk Constraint,” *Scandinavian Actuarial Journal*, 2020, 419–453.
- YAARI, M. (1987): “The Dual Theory of Choice under Risk,” *Econometrica*, 55, 95–115.
- YANG, L., C. ZHANG, AND H. ZHU (2022): “Robust Stochastic Stackelberg Differential Reinsurance and Investment Games for an Insurer and a Reinsurer with Delay,” *Methodology and Computing in Applied Probability*, 24, 361–384.
- ZHU, M. B., M. GHOSSOUB, AND T. J. BOONEN (2023): “Equilibria and Efficiency in a Reinsurance Market,” *Insurance: Mathematics and Economics*, 113, 24–49.

Maria Andraos: UNIVERSITY OF WATERLOO – DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE – 200 UNIVERSITY AVE. W. – WATERLOO, ON, N2L 3G1 – CANADA

Email address: mandraos@uwaterloo.ca

Mario Ghossoub: UNIVERSITY OF WATERLOO – DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE – 200 UNIVERSITY AVE. W. – WATERLOO, ON, N2L 3G1 – CANADA

Email address: mario.ghossoub@uwaterloo.ca

Bin Li: UNIVERSITY OF WATERLOO – DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE – 200 UNIVERSITY AVE. W. – WATERLOO, ON, N2L 3G1 – CANADA

Email address: bin.li@uwaterloo.ca

Benxuan Shi: UNIVERSITY OF WATERLOO – DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCE – 200 UNIVERSITY AVE. W. – WATERLOO, ON, N2L 3G1 – CANADA

Email address: benxuan.shi1@uwaterloo.ca